



RKDF UNIVERSITY RANCHI

Credit System and Marks Distribution:-

Semester-I

SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	BSC	DE101	Applied Chemistry	3	0	0	3	30	70	21	100	35
2	BSC	DE102	Mathematics - I	3	0	0	3	30	70	21	100	35
3	Huminties & Social Sc. Courses	DE103	Communication Skills in English	2	0	0	2	30	70	21	100	35
4	BSC	DE104	Applied Physics	3	0	0	3	30	70	21	100	35
5	ESC	DE105	Fundamentals of Computer	2	0	0	2	30	70	21	100	35
6	ESC	DE106	Engineering Graphics	2	0	0	2	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	BSC	DE151	Applied Chemistry Lab	0	0	2	1	30	20		50	25
2	Huminties & Social Sc. Course	DE153	Communication Skills in English	0	0	2	1	30	20		50	25
3	BSC	DE154	Applied Physics Lab	0	0	2	1	30	20		50	25
4	ESC	DE155	Fundamentals of Computer Lab	0	0	2	1	30	20		50	25
5	ESC	DE156	Engineering Graphics Lab	0	0	3	1.5	30	20		50	25
6	ESC	DE157	Workshop / Manufacturing Practices Lab	0	0	3	1.5	30	20		50	25
TOTAL							22					



RKDF UNIVERSITY RANCHI

Credit System and Marks Distribution :-

Semester-II												
SL. No.	Category	Subject Code	Subject Name	Periods			Credits	Marks Distribution				
				L	T	P		Internal	External		Total	
								Max	Max	Min	Max	Min
1	BSC	DE201	Mathematics – II	3	0	0	3	30	70	21	100	35
2	BSC	DE202	Applied Physics – II	3	0	0	3	30	70	21	100	35
3	ESC	DE203	Fundamentals of Electrical & Electronics Engineering	2	0	0	2	30	70	21	100	35
4	ESC	DE204	Engineering Mechanics	2	0	0	2	30	70	21	100	35
5	Huminitie s & Social Sc. Course	DE205	Communication Skills in EnglishII	3	0	0	3	30	70	21	100	35
PRACTICAL DEMONSTRATION												
1	BSC	DE252	Applied Physics –II Lab	0	0	2	1				50	25
2	ESC	DE253	Fundamentals of Electrical & Electronics Engineering Lab	0	0	2	1				50	25
3	ESC	DE254	Engineering Mechanics Lab	0	0	2	1				50	25
TOTAL							16					



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE101
Course Title	Applied Chemistry
Credits	3 (L: 3 T: 0 P: 0)

Course Objectives:

1. The students will acquire a foundation in chemistry of sufficient breadth and depth to enable them to understand and critically interpret the primary chemical literature.
2. The students will develop the ability to effectively communicate scientific information and research results in written and oral formats
3. The students will learn professionalism, including the ability to work in teams and apply basic ethical principles.

UNIT 1

Atomic Structure and Chemical Bonding:

Atomic Structure: Definition of atom, Fundamental particles of atom –electron, neutron, protons, Definition of Atomic no, Atomic mass no Isotopes & Isobars & their distinction with suitable examples, Bohr's and

Rutherford's theory; Definition and Shape of the orbital & distinction between orbits and orbitals, Hund's Rule, filling up the orbital's by Aufbau's principle (till Atomic no. 30) Chemical bonding: Cause of chemical bonding, types of bonds: electrovalent, covalent and coordinate bonds, formation of electrovalent & covalent compounds Eg. NaCl, CaCl₂, CO₂, Cl₂, NH₃, C₂H₄, N₂, C₂H₂, etc., coordination bond in NH₄⁺, and anomalous properties of NH₃, H₂O due to hydrogen bonding, and metallic bonding, Octet rule, Duplet rule.

UNIT 2

Water and Corrosion:

Water: Characteristics, Sources, Impurities, Hard & Soft Water, Causes of Hardness, Types of Hardness, Degree of Hardness, Boiler and Steam Generation, Scale & Sludge Formation – Causes, Disadvantage, Softening Methods such as Boiling, Clark's, Soda Ash, Lime Soda, Zeolite & Ion Exchange Methods with Principle Chemical Reactions. Plumbo solvency & its Removal. Numerical Problems. Corrosion: Definition of Corrosion, Types of Corrosion (Dry and Wet chemical Corrosion) and their mechanism. Protection of metal from corrosion (Corrosion Control). Application of Protective Coatings like metal coating such as Galvanizing, Tinning, Metal Spraying, Sherardizing, Electroplating and Metal Cladding.

UNIT 3

Electrochemistry and Electrochemical Cells:

Electrochemistry: Definition of terms: Conductors, Insulators, Dielectrics, Electrolyte, Non-Electrolyte, Electrolysis, Electrolytic Cell with suitable examples, Electrolytic dissociation, Arrhenius Theory of Ionization, Degree of Ionization & factors affecting degree of ionization. Redox reactions. Faradays laws of electrolysis and simple numerical problems. Electrochemical cells: Concept of electrode potential such as reduction potential & oxidation potential. Electrochemical Series, Electrolysis of CuSO₄ Solution by using Cu Electrode & Platinum Electrode, Electrochemical Cells & Batteries, Definition, types such as Primary & Secondary Cells & their examples.

UNIT 4

Lubricants and Fuels:

Lubricants: Definition, Classification with examples. Functions of lubricant, Lubrication - Mechanism of Lubrication (Fluid Film, Boundary and Extreme Pressure). Physical Characteristics of Lubricants Such as Viscosity, Viscosity Index, Oiliness, Volatility, Flash & Fire Point, and Cloud & Pour Point, Chemical Characteristics such as Acid Value or Neutralization Number, Emulsification, Saponification Value, Selection of Lubricants, Characteristics of Transformer oil Fuels: Definition and combustion of fuel, classification of fuels, calorific values (HCV and LCV), calculation of HCV and LCV using Dulong's formula. Proximate analysis of coal solid fuel petrol and diesel - fuel rating (octane and cetane numbers), Chemical composition, calorific values and applications of LPG, CNG, water gas, coal gas, producer gas and biogas.

UNIT 5

Metals and Non - Metallic Materials:

Metals: Occurrence of Metal such as Iron, Aluminum, Chromium, Nickel, Tin, their properties Definition of Metallurgy Mineral, Ore Gangue Flux & Slag, stages of Extraction of metal from Its Ores in detail. Alloys: definition of alloy, purposes of making alloy. preparation methods, General Principal of metallurgy, minerals/ ores, ore dressing, roasting, smelting, bessemerization, fluxes, purification. Explanation of alloying purposes, methods of preparation, composition and uses of alloy like brass, bronze, duralumin, German silver, gun metal, solder, stainless steel, casting and bearing alloy. Nonmetallic materials: Definition of Polymers, formation of Polymers by Addition & condensation polymerization. Properties and uses of PVC, polyethene, polystyrene, polyamides, polyesters, Bakelite. Synthetic fibers – nylon, rayon, decron, and polyesters. Natural Rubber rubber its processing and drawbacks, vulcanization of rubber with chemical reaction, synthetic rubber, definition & distinction between natural & synthetic rubber Thermal insulating material : definition & characteristics of thermal insulator. Preparation, properties & application of thermocol & glass wool Properties & application of Asbestos cork

Text Books:

1. Text Book of Chemistry for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi, 2017-18. TEXT

Reference Books

1. Engineering Chemistry Jain & Jain, Dhanpat Rai and Sons; New Delhi, 2015
2. Engineering Chemistry S. S. Dara & Dr.S.S.Umare, S. Chand Publication; New Delhi, 2015.
3. Industrial Chemistry B. K. Sharma, Goel Publication

4. Engineering Chemistry By O P Agarwal (Khanna Publication)
5. Environmental Chemistry & Pollution Control S. S. Dara, S. Chand Publication
6. Chemistry for Engineers Agnihotri, Rajesh, Wiley India Pvt. Ltd., 2014.
7. Engineering Chemistry Rao and Agarwal
8. Engineering Chemistry P.C. Jain
9. Applied Chemistry H.N. Sahni
10. Polytechnic Chemistry Vedprakash Mehta, Jain brothers.
11. Engineering Chemistry Uppal

Course Outcomes:

1. Students will be able to do when they successfully complete a learning experience whether it is project, course or program
- Typically, we divide their course into smaller units such. Modules, mapwork, flowchart etc

As a general rule applied, as the level of analysis becomes smaller, from courses to module to assignment. Students will be able to function as a member of an interdisciplinary problems solving team

Students will appreciate the central role of chemistry in our society and we use this as a basis for ethical behavior in issues facing chemists including an understanding of safe handling of chemical, environmental issues and key issues facing our society in energy health and medicine.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE151
Course Title	Applied Chemistry Lab
Credits	1 (L: 0 T: 0 P: 2)

List of Experiments:

1. Determination of Total hardness by EDTA method.
2. Determination of Total hardness by Clarke's method.
3. Determination of Flash & Fire Points by Pensky Marten Apparatus.
4. Determination of Flash & Fire Points by Abel's Apparatus.
5. Determination of Viscosity and Viscosity index by Redwood viscometer No.1.
6. Determination of Viscosity and Viscosity index by Redwood viscometer No.2.
7. Determination of percentage of Copper in Brass by Iodometric Titration.
8. To prepare a solution of N/20 sodium carbonates and find the strength of HCl using N/20 sodium carbonate solution.

Reference Books:

1. Applied Chemistry Laboratory Practices Vol. I and Vol. II, Dr. G. H. Hugar & Prof A. N. Pathak ,NITTTR, Chandigarh, Publications, 2013-14. LAB



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE102
Course Title	Mathematics-I
Credits	3 (L: 3 T: 0 P: 0)

UNIT-I

Trigonometry: Concept of angles, measurement of angles in degrees, grades and radians and their conversions, T-Ratios of Allied angles (without proof), Sum, difference formulae and their applications (without proof). Product formulae (Transformation of product to sum, difference and vice versa). T- Ratios of multiple angles, sub-multiple angles ($2A$, $3A$, $A/2$). Graphs of $\sin x$, $\cos x$, $\tan x$ and e^x

UNIT – II

Differential Calculus: Definition of function; Concept of limits. Four standard limits:

Differentiation by definition of $\sin x$ and $\cos x$, $\tan x$, $\log x$ Differentiation of sum, product and quotient of functions. Differentiation of function of a function. Differentiation of trigonometric and inverse trigonometric functions, Logarithmic differentiation, Exponential functions.

UNIT-III

Algebra: Complex Numbers: Definition, real and imaginary parts of a Complex number, polar and Cartesian, representation of a complex number and its conversion from one form to other, conjugate of a complex number, modulus and amplitude of a complex number Addition, Subtraction, Multiplication and Division of a complex number. De-Moivre's Theorem, its application.

UNIT – IV

Partial fractions: Definition of polynomial fraction proper & improper fractions and definition of partial fractions. To resolve proper fraction into partial fraction with denominator containing non-repeated linear factors, repeated linear factors and irreducible non-repeated quadratic factors. To resolve improper fraction into partial fraction

UNIT – V

Permutations and Combinations: Value of nPr and nCr

Binomial theorem: Binomial theorem (without proof) for positive integral index (expansion and general form); binomial theorem for any index (expansion without proof) first and second binomial approximation with applications to engineering problems

Reference Books:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 40th Edition, 2007.
2. G. B. Thomas, R. L. Finney, Calculus and Analytic Geometry, Addison Wesley, 9th Edition, 1995.
3. Reena Garg, Engineering Mathematics, Khanna Publishing House, New Delhi (Revised Ed. 2018)
4. V. Sundaram, R. Balasubramanian, K.A. Lakshminarayanan, Engineering Mathematics, 6/e., Vikas Publishing House.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE103
Course Title	Communication skills in English
Credits	2 (L: 2 T: 0 P: 0)

UNIT -1

Application of Grammar, Verbs Tense. Do as directed (active/passive, Direct/ Indirect, affirmative/ Negative/ Assertive/ Interrogative, Question tag, remove too, use of article, preposition, conjunction, punctuation). Correct the errors from the sentences. Vocabulary Building (Synonyms/ Antonyms/ Homophones/ Use of contextual word in given paragraph)

UNIT-2

Introduction to communication Definition, Communication cycle Concept of Communication Process Formal Communication Formal: Types- a) Vertical Communication b) Horizontal Communication Informal: Types- Diagonal Communication Verbal Vs Non-Verbal Communication.

Verbal: Types a) Oral Communication b) Written Communication Non-Verbal: Types a) Body Language b) Graphic Language

UNIT-3

Principal of Effective Communication. Principal of Effective Communication, Communication barriers and how to overcome them.

UNIT-4

Developing Effective message: Thinking about purpose, knowing the audience, structuring the message, selecting proper channels, minimize barriers and facilitating feedback. (Example: Writing articles for newspapers, magazines)

Suggested books:

1. Contemporary English Grammar Structures and Composition; David Green, Macmillan
2. English Grammar and composition; R. C. Jain, Macmillan
3. Effective Technical Communication; M. Ashraf Rizvi, Tata McGraw Hill Companies
4. Developing communication Skills; Krushna Mohan, Meera Baneji, Macmillan.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE104
Course Title	Applied physics-I
Credits	3 (L: 3 T: 0 P: 0)

Course Objectives:

Couse objectives- study of applied physics aims to give an understanding of physical world by observation and predictions. The course help diploma engineers to apply the basic concepts and principles to solve engineering problems and to understand different technology-based application.

Unit -1

Physical world, Units and Measurements: Physical quantities; fundamental and derived, Units and systems of units (FPS, CGS and SI units), Dimensions and dimensional formulae of physical quantities, Principle of homogeneity of dimensions, Dimensional equations and their applications (conversion from one system of units to other, checking of dimensional equations and derivation of simple equations), Limitations of dimensional analysis. Measurements: Need, measuring instruments, least count, types of measurement (direct, indirect), Errors in measurements (systematic and random), absolute error, relative error, error propagation, error estimation and significant figures.

Unit-2

Force and Motion: Scalar and Vector quantities – examples, representation of vector, types of vectors. Addition and Subtraction of Vectors, Triangle and Parallelogram law (Statement only), Scalar and Vector Product, Resolution of a Vector and its application to inclined plane and lawn roller. Force, Momentum, Statement and derivation of conservation of linear momentum, its applications such as recoil of gun, rockets, Impulse and its applications. Circular motion, definition of angular displacement, angular velocity, angular acceleration, frequency, time period, Relation between linear and angular velocity, linear acceleration and angular acceleration (related numerical), Centripetal and Centrifugal forces with live examples, Expression and applications such as banking of roads and bending of cyclist.

Unit-3

Work, Power and Energy: Work Concept and units, examples of zero work, positive work and negative work Friction: concept, types, laws of limiting friction, coefficient of friction, reducing friction and its engineering applications, Work done in moving an object on horizontal and inclined plane for rough and plane surfaces and related applications. Energy and its units, kinetic energy, gravitational potential energy with examples and derivations, mechanical energy, conservation of mechanical energy for freely falling bodies, transformation of energy (examples). Power and its

units, power and work relationship, calculation of power (numerical problems)

Unit-4

Rotational Motion: Translational and rotational motions with examples, Definition of torque and angular momentum and their examples, Conservation of angular momentum (quantitative) and its applications. Moment of inertia and its physical significance, radius of gyration for rigid body, theorems of parallel and perpendicular axes (statements only), Moment of inertia of rod, disc, ring and sphere (hollow and solid); (Formulae only).

Unit-5

Properties of Matter: Elasticity: definition of stress and strain, moduli of elasticity, Hooke's law, significance of stress-strain curve. Pressure: definition, units, atmospheric pressure, gauge pressure, absolute pressure, Fortin's Barometer and its applications. Surface tension: concept, units, cohesive and adhesive forces, angle of contact, Ascent Formula (No derivation), applications of surface tension, effect of temperature and impurity on surface tension. Viscosity and coefficient of viscosity: Terminal velocity, Stoke's law and effect of temperature on viscosity, application in hydraulic systems. Hydrodynamics: Fluid motion, stream line and turbulent flow, Reynold's number Equation of Continuity, Bernoulli's Theorem (only formula and numerical) and its applications.

Heat and Thermometry: Concept of heat and temperature, modes of heat transfer (conduction, convection and radiation with examples), specific heats, scales of temperature and their relationship, Types of Thermometers (Mercury thermometer, Bimetallic thermometer, Platinum resistance thermometer, Pyrometer) and their uses. Expansion of solids, liquids and gases, coefficient of linear, surface and cubical expansions and relation amongst them, Co-efficient of thermal conductivity, engineering applications.

Text Books:

1. Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi

Reference Books:

1. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi.
2. Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi
3. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi
4. Engineering Physics by DK Bhattacharya & Poonam Tandan; Oxford University Press, New Delhi

Course Outcomes:

After undergoing this subject, the students will be able to-

1. Identify physical quantity, select their units for engineering solutions and make measurement with accuracy.

2. Describe the different forms of energy, methods of transfer of energy.
3. Represent the physical quantity as scalar and vectors to solve real life relevant problems.
4. Analyze the type of motion and apply the formulation to understand banking pf road and conservation of momentum, recoil of gun etc.
5. Describe the concept of work, energy and power with their units and derive the relationship for work energy and power.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE154
Course Title	Applied Physics-I Lab
Credits	1 (L: 0 T: 0 P: 2)

Course objectives-

The course aims to supplement the factual knowledge gained in the lecture by first hand manipulation of apparatus. This will develop scientific temper and help student to apply the basic concepts in solving engineering and technology-based problems. In addition, students get confidence in handling equal and thus learn skill of measurements.

List of Practical:

- 1) To measure the length, radius of a given cylinder, a test tube using vernier caliper and find the volume of each object.
- 2) To determine the diameter of a wire, a solid ball using screw gauge. 3) To Verify the Ohm's law
- 4) To verify the Kirchhoff's law (kvl and kcl)
- 5) To find the surface tension of a liquid by capillary rise method.
- 6) To determine the viscosity of a given liquid (Glycerin) by Stoke's law.
- 7) To verify the law of conservation of mechanical energy.
- 8) To measure the room temperature of hot bath using mercury Thermometer and convert it into different scale.
- 9) To determine the radius of curvature of concave and convex mirror using spherometer.
- 10) Determine the force constant using Hook's law.

Course Outcomes:

After undergoing this lab work, students will be able to -

1. Select the right kind of measuring tools (meter scale, screw gauge, vernier calipers etc)
2. Describe and verify the Hook's law and determine the force constant of a spring body.
3. Understand the Stoke's law and determine the viscosity of a given liquid.
4. Understand the use of thermometer to measure the temperature under different conditions and scales.
5. Understand the current and voltage relationship.

Reference Books:

1. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi.
2. Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi
3. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi
4. Engineering Physics by DK Bhattacharya & Poonam Tandan; Oxford University Press, New Delhi



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE105
Course Title	Fundamentals of Computer
Credits	2(L: 2 T: 0 P: 0)

Unit- 1

Basic Internet skills: Understanding browser, efficient use of search engines, awareness about Digital India portals (state and national portals) and college portals. General understanding of various computer hardware Components — CPU, Memory, Display, Keyboard, Mouse, HDD and other Peripheral Devices.

Unit- 2

OS Installation (Linux and MS Windows), Unix Shell and Commands, vi editor.
HTML4, CSS, making basic personal webpage.

Unit- 3

Unit- 4

Office Tools: Open Office Writer, Open Office Spreadsheet (Calc), Open Office Impress.

Unit-5

Information security best practices. Class lectures will only introduce the topic or demonstrate tool, actual learning will take place in the Lab by practicing



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE106
Course Title	Engineering Graphics
Credits	2(L: 2 T: 0 P: 0)

UNIT – I

Basic elements of Drawing: Drawing Instruments and supporting materials: method to use them with applications. Convention of lines and their applications. Representative Fractions – reduced, enlarged and full size scales; Engineering Scales such as plain and diagonal scale. Dimensioning techniques as per SP-46:2003 – types and applications of chain, parallel and coordinate dimensioning. Geometrical and Tangency constructions. (Redraw the figure)

UNIT – II

Orthographic projections: Introduction of projections-orthographic, perspective, isometric and oblique: concept and applications. (No question to be asked in examination). Introduction to orthographic projection, First angle and Third angle method, their symbols. Conversion of pictorial view into Orthographic Views – object containing plain surfaces, slanting surfaces, slots, ribs, cylindrical surfaces. (use First Angle Projection method only)

UNIT – III

Isometric Projections: Introduction to isometric projections. Isometric scale and Natural scale. Isometric view and isometric projection. Illustrative problems related to objects containing lines, circles and arcs shape only. Conversion of orthographic views into isometric view/projection.

UNIT – IV

Free Hand Sketches of engineering elements: Free hand sketches of machine elements: Thread profiles, nuts, bolts, studs, set screws, washer, locking arrangements. (For branches other than mechanical Engineering, the teacher should select branch specific elements for free hand sketching) Free hand sketches of orthographic view (on squared graph paper) and isometric view (on isometric grid paper)

UNIT – V

Computer aided drafting interface: Computer Aided Drafting: concept. Hardware and various CAD software available. System requirements and understanding the interface. Components of AutoCAD software window: Title bar, standard tool bar, menu bar, object properties tool bar, draw tool bar, modify tool bar, cursor cross hair. Command window, status bar, drawing area, UCS icon.

File features: new file, Saving the file, opening an existing drawing file, Creating templates, Quit. Setting up new drawing: Units, Limits, Grid, Snap. Undoing and redoing action.

Unit – VI

Computer aided drafting: Draw basic entities like Line, Circle, Arc, Polygon, Ellipse, Rectangle, Multiline, Poly Line. Method of Specifying points: Absolute coordinates, Relative Cartesian and Polar coordinates. Modify and edit commands like trim, extend, delete, copy, offset, array, block, layers. Dimensioning: Linear, Horizontal Vertical, Aligned, Rotated, Baseline, Continuous, Diameter, Radius, Angular Dimensions. Dim scale variable.

Editing dimensions. Text: Single line Text, Multiline text. Standard sizes of sheet. Selecting Various plotting parameters such as Paper size, paper units, Drawing orientation, plot scale, plot offset, plot area, printpreview.

Reference Books:

1. Bhatt, N. D. *Engineering Drawing*, Charotar Publishing House, Anand, Gujrat 2010; ISBN: 978-93-80358-17-8
2. Dhawan, R. K. *Engineering Drawing*. S. Chand and Company, New Delhi; ISBN: 81-219-1431-0.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-I
(Common to all Branches)

Course Code	DE157
Course Title	Workshop/Manufacturing Practices Lab
Credits	1.5(L: 0 T: 0 P: 3)

Details of Practical Content:

1. Carpentry:

- (i) Demonstration of different wood working tools / machines.
- (ii) Demonstration of different wood working processes, like planing, marking, chiseling, grooving, turning of wood etc.
- (iii) One simple job involving any one joint like mortise and tenon dovetail, bridle, half lap etc.

2. Fitting:

- (i) Demonstration of different fitting tools and drilling machines and power tools
- (ii) Demonstration of different operations like chipping, filing, drilling, tapping, sawing, cutting etc.
- (iii) One simple fitting job involving practice of chipping, filing, drilling, tapping, cutting etc.

3. Welding:

- (i) Demonstration of different welding tools / machines.
- (ii) Demonstration on Arc Welding, Gas Welding, MIG, MAG welding, gas cutting and rebuilding of broken parts with welding.
- (iii) One simple job involving butt and lap joint.

4. Sheet Metal Working:

- (i) Demonstration of different sheet metal tools / machines.
- (ii) Demonstration of different sheet metal operations like sheet cutting, bending, edging, end curling, lancing, soldering, brazing, and riveting.
- (iii) One simple job involving sheet metal operations and soldering and riveting.

5. Electrical House Wiring:

- (i) Practice on simple lamp circuits
- (ii) One lamp controlled by one switch by surface conduit wiring,
- (iii) Lamp circuits- connection of lamp and socket by separate switches,
- (iv) Connection of Fluorescent lamp/tube light,
- (v) simple lamp circuits-install bedroom lighting.
- (vi) Simple lamp circuits- install stair case wiring.

6. Demonstration:

- (i) Demonstration of measurement of Current, Voltage, Power and Energy.
- (ii) Demonstration of advance power tools, pneumatic tools, electrical wiring tools and accessories.
- (iii) Tools for Cutting and drilling

References:

1. S.K. Hajara Chaudhary, Workshop Technology, Media Promoters and Publishers, New Delhi, 2015
2. B.S. Raghuwanshi, Workshop Technology, Dhanpat Rai and sons, New Delhi 2014
3. K. Venkat Reddy, Workshop Practice Manual, BS Publications, Hyderabad



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-II
(Common to all Branches)

Course Code	DE202
Course Title	Applied Physics-II
Credits	3(L: 3 T: 0 P: 0)

Course Objectives:

Applied physics aims to give an understanding of this world both by observation and prediction of the way in which the object behaves; this course helps the student to solve board-based engineering problems.

Unit – 1

Wave motion and its applications:

Wave motion, transverse and longitudinal waves with examples, definitions of wave velocity, frequency and wave length and their relationship, Sound and light waves and their properties, Wave equation, Amplitude, Phase, Phase difference, principle of superposition of waves and beat formation. Simple Harmonic Motion (SHM): definition, expression for displacement, velocity, acceleration, time period, frequency etc. Simple harmonic progressive wave and energy transfer, study of vibration of cantilever and determination of its time period, Free, forced and resonant vibrations with examples.

Acoustics of buildings – reverberation, reverberation time, echo, noise, coefficient of absorption of sound, methods to control reverberation time and their applications, Ultrasonic waves – Introduction and properties, engineering and medical applications of ultrasonic.

Unit – 2

Optics: Basic optical laws; reflection and refraction, refractive index, Images and image formation by mirrors, lens and thin lenses, lens formula, power of lens, magnification and defects. Total internal reflection, Critical angle and conditions for total internal reflection, applications of total internal reflection in optical fiber. Optical Instruments; simple and compound microscope, astronomical telescope in normal adjustment, magnifying power, resolving power, uses of microscope and telescope, optical projection systems.

Unit – 3

Electrostatics: Coulombs law, unit of charge, Electric field, Electric lines of force and their properties, Electric flux, Electric potential and potential difference, Gauss law: Application of Gauss law to find electric field intensity of straight charged conductor, plane charged sheet and charged sphere. Capacitor and its working, Types of capacitors, Capacitance and its units. Capacitance of a parallel plate capacitor, Series and parallel combination of capacitors (related numerical), dielectric and its effect on capacitance, dielectric break down.

Unit – 4

Current Electricity: Electric Current and its units, Direct and alternating current, Resistance and

its units, Specific resistance, Conductance, Specific conductance, Series and parallel combination of resistances. Factors affecting resistance of a wire, carbon resistances and colour coding. Ohm's law and its verification, Kirchhoff's laws, Wheatstone bridge and its applications (slide wire bridge only), Concept of terminal potential difference and Electro motive force (EMF) Heating effect of current, Electric power, Electric energy and its units (related numerical problems), Advantages of Electric Energy over other forms of energy.

Unit – 5

Electromagnetism: Types of magnetic materials; dia, para and ferromagnetic with their properties, Magnetic field and its units, magnetic intensity, magnetic lines of force, magnetic flux and units, magnetization. Concept of electromagnetic induction, Faraday's Laws, Lorentz force (force on moving charge in magnetic field). Force on current carrying conductor, force on rectangular coil placed in magnetic field. Moving coil galvanometer; principle, construction and working, Conversion of a galvanometer into ammeter and voltmeter.

Semiconductor Physics: Energy bands in solids, Types of materials (insulator, semi-conductor, conductor), intrinsic and extrinsic semiconductors, p-n junction, junction diode and V-I characteristics, types of junction diodes. Diode as rectifier – half wave and full wave rectifier (centre tapped). Transistor; description and three terminals, Types- pnp and npn, some electronic applications (list only). Photocells, Solar cells; working principle and engineering applications.

Text Books:

Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi

Reference Books:

Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi

Course Outcomes:

After undergoing through syllabus, the students will be able to

- 1) Explain the wave motion, simple harmonic motion, polarization of wave etc.
- 2) Explain the ultrasonic waves and engineering, medical and industrial application of ultra sound.
- 3) Apply the knowledge of diodes, led, power adapters in electronic circuit.
- 4) Explain the current as flow of charge, concept power and resistance. Etc.
- 5) Express the nano science and technique and their effects on nanosecond and science and their impact on society. Etc.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-II
(Common to all Branches)

Course Code	DE203
Course Title	Fundamental of Electrical & Electronics Engineering
Credits	3(L: 3 T: 0 P: 0)

Course Objectives:

1. To familiarize students with basic electrical and electronic components and their functions
2. To make the students aware with AC and DC circuits and their analysis.

Unit-I

Overview of Electronic Components & Signals: Passive Active Components: Resistances, Capacitors, Inductors, Signals: DC/AC, voltage/current, periodic/non-periodic signals, average, rms, peak values, different types of signal waveforms, Ideal/non-ideal voltage/current sources, independent/dependent voltage current sources.

Unit- II

Diodes, Transistors, FET, MOS and CMOS and their Applications, Overview of Digital Electronics: Introduction to Boolean Algebra, Electronic Implementation of Boolean Operations, OP AMP 741

Unit- III

Electric and Magnetic Circuits: EMF, Current, Potential Difference, Power and Energy; M.M.F, magnetic force, permeability, hysteresis loop, reluctance, leakage factor and BH curve; Electromagnetic induction, Faraday's laws of electromagnetic induction, Lenz's law

Unit-IV

A.C. Circuits: Cycle, Frequency, Periodic time, Amplitude, Angular velocity, RMS value, Average value, Form Factor Peak Factor, impedance, phase angle, and power factor; Mathematical and phasor representation of alternating emf and current; Voltage and Current relationship in Star and Delta connections; A.C in resistors, inductors and capacitors; A.C in R-L series, R-C series, R-L-C series and parallel circuits; Power in A. C. Circuits, power triangle, Introduction Overview and applications of Electrical Machines DC Motor, DC Generator, AC Generator, Induction Motor and Transformer.

Text Books:

1. Basic Electrical Engineering I.J. Nagrath and D.P. Kothari, 2nd Edition, TMH, Delhi.
2. Basic Electrical Engineering by UA Bakshi and AV Bakshi Technical Publications Pune.

Reference Books:

1. Ritu Sahdev, Basic Electrical Engineering, Khanna Publishing House
2. Mittle and Mittal, Basic Electrical Engineering, McGraw Education, New Delhi, 2015, ISBN:978-0-07-0088572-5
3. Saxena, S. B. Lal, Fundamentals of Electrical Engineering, Cambridge University Press, latest

edition ISBN: 9781107464353

Course Outcomes:

After completing the course students will be able to

1. Discuss about basic electrical and electronic devices
2. Solve network problems of AC and DC circuits
3. Can design basic analog and digital circuits.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-II
(Common to all Branches)

Course Code	DE253
Course Title	Fundamental of Electrical & Electronics Engineering Lab
Credits	1(L: 0 T: 0 P: 2)

Course Objectives:

1. To make the students able to do design and make connection for basic electrical circuit.
2. To enable students to verify the characteristics of the electrical and electronic circuits practically
3. To have sound practice of electrical safety.

List of Experiments

1. To study the various components of Electrical and Electronics.
2. To determine the stated value of a resistor by interpreting the color code indicated on the resistor.
3. To measure resistance using ohmmeter
4. To study and verify the series and parallel Circuit.
5. To study and plot V-I Characteristics of Silicon P-N Junction Diode.
6. To study the dependence of current on the potential difference across a resistor and determine its resistance. Also plot a graph between V and I.
7. To study the Inverting Amplifier and Non-Inverting Amplifier op-amp circuits.
8. To study the basic logic gates: AND, OR, NOT
9. To Study the Characteristics of Half – Wave Rectifier.
10. To study the Characteristics of Light Emitting Diode

Course Outcomes:

1. Students will be able to analyze and design electrical and electronic circuits practically.
2. Students will be able to study the characteristics of basic electrical circuits.
3. Students will be able to follow electrical safety in all prospects.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-II
(Common to all Branches)

Course Code	DE204
Course Title	Engineering Mechanics
Credits	3(L: 3 T: 0 P: 0)

Course Objectives:

Following are the objectives of this course:

1. To obtain resultant of various forces
2. To calculate support reactions through conditions of equilibrium for various structures
3. To understand role of friction in equilibrium problems
4. To know fundamental laws of machines and their applications to various engineering problems

Unit -I

Basics of mechanics and force system: Significance and relevance of Mechanics, Applied mechanics, Statics, Dynamics. Space, time, mass, particle, flexible body and rigid body. Scalar and vector quantity, Units of measurement (SI units) - Fundamental units and derived units. Force – unit, representation as a vector and by Bow's notation, characteristics and effects of a force, Principle of transmissibility of force, Force system and its classification. Resolution of a force - Orthogonal components of a force, moment of a force, Varignon's Theorem. Composition of forces – Resultant, analytical method for determination of resultant for concurrent, non-concurrent and parallel co-planar force systems – Law of triangle, parallelogram and polygon of forces.

Unit- II

Equilibrium: Equilibrium and Equilibrant, Free body and Free body diagram, Analytical and graphical methods of analyzing equilibrium Lami's Theorem – statement and explanation, Application for various engineering problems. Types of beams, supports (simple, hinged, roller and fixed) and loads acting on beam (vertical and inclined point load, uniformly distributed load, couple), Beam reaction for cantilever, simply supported beam with or without overhang – subjected to combination of Point load and uniformly distributed load. Beam reaction graphically for simply supported beam subjected to vertical point loads only.

Unit- III

Friction: Friction and its relevance in engineering, types and laws of friction, limiting equilibrium, limiting

friction, co-efficient of friction, angle of friction, angle of repose, relation between co-efficient of friction and angle of friction. Equilibrium of bodies on level surface subjected to force parallel and inclined to plane.

Equilibrium of bodies on inclined plane subjected to force parallel to the plane only.

Unit- IV

Centroid and center of gravity: Centroid of geometrical plane figures (square, rectangle, triangle, circle, semi-circle, quarter circle) Centroid of composite figures composed of not more than three

geometrical figures Centre of Gravity of simple solids (Cube, cuboid, cone, cylinder, sphere, hemisphere) Centre of Gravity of composite solids composed of not more than two simple solids.

Unit – V

Simple lifting machine: Simple lifting machine, load, effort, mechanical advantage, applications and advantages. Velocity ratio, efficiency of machines, law of machine. Ideal machine, friction in machine, maximum Mechanical advantage and efficiency, reversible and non-reversible machines, conditions for reversibility Velocity ratios of Simple axle and wheel, Differential axle and wheel, Worm and worm wheel, Single purchase and double purchase crab winch, Simple screw jack, Weston's differential pulley block, geared pulley block.

Reference Books:

1. D.S. Bedi, Engineering Mechanics, Khanna Publications, New Delhi (2008)
2. Khurmi, R.S., Applied Mechanics, S. Chand & Co. New Delhi.
3. Bansal R K, A text book of Engineering Mechanics, Laxmi Publications.
4. Ramamrutham, Engineering Mechanics, S. Chand & Co. New Delhi.

Course outcomes:

After completing this course, student will be able to:

1. Identify the force systems for given conditions by applying the basics of mechanics.
2. Determine unknown force(s) of different engineering systems.
3. Apply the principles of friction in various conditions for useful purposes.
4. Find the centroid and centre of gravity of various components in engineering systems.
5. Select the relevant simple lifting machine(s) for given purposes.



R.K.D.F. UNIVERSITY, RANCHI
Diploma Engineering Semester-II
(Common to all Branches)

Course Code	DE205
Course Title	Communication Skills-2
Credits	3(L: 3T: 0 P: 0)

Course objectives:

UNIT -1

Formal written skills Office Drafting: Circular, Notice and Memo Job Application with resume.
Business correspondence: Enquiry, Order letter, Complaint letter and Adjustment letter.
Report writing: Accident report, fall in production, Progress report, Investigation report.

UNIT -2

Principles of Effective Communication Principles of Effective listening/Speaking Communication
Barriers, Overcoming Barrier

UNIT -3

Soft skill development

Speaking skill

Introduction to Group discussion Process of Group Discussion Leadership skill

Instant public speaking

UNIT -4

Etiquettes and Body Language Telephone etiquettes listening/Speaking Problems of telephonic conversation Verbal /oral etiquettes

Physical appearance

Eye contact/ Body Language Group Discussion

Suggested Books:

2. Contemporary English Grammar Structures and Composition; David Green, Macmillan
3. English Grammar and composition; R. C. Jain, Macmillan
4. Effective Technical Communication; M. Ashraf Rizvi, Tata McGraw Hill Companies
5. Developing Communication Skills; Krushna Mohan, Meera Baneji, Macmillan



DIPLOMA IN MINING ENGINEERING												
SEMESTER III												
S. No	Subject code	Name of Subject	Period			Evaluation Scheme				Subject	Credit	Hours
			L	T	P	Assignment	T A	Total	ESC			
1	8D.201	Mechanical Engineering	3	0	0	20	10	30	70	100	3	3
2	8D.202	Mining Geology I	3	0	0	20	10	30	70	100	3	3
3	8D.203	Mine Surveying I	3	0	0	20	10	30	70	100	3	3
4	8D.204	Introduction to Mining	3	0	0	20	10	30	70	100	3	3
5	9D.204	Applied mathematics	3	0	0	20	10	30	70	100	3	3
6	40D.201	Communication & Soft Skills	2	0	0	20	10	30	70	100	2	2
7	14D.201	** Disaster Management	3	0	0	20	10	30	70	100	0	3
8	40D.101	***Life Skills I	2	0	0	20	10	30	70	100	0	2
PRACTICAL / SESSIONAL												
1	8DP.201	Mechanical Engineering Lab	-	-	2	-	30	30	20	50	1	2
2	8DP.202	Mining Geology I Lab	-	-	2	-	30	30	20	50	1	2
3	8DP.203	Mine Surveying I Lab	-	-	2	-	30	30	20	50	1	2
								Total		750	20	23

- **** NOTE: Qualifying Non Credit Course**
- ***** NOTE: For Lateral students (Qualifying Non Credit Course)**
- **Note: Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of IIIrd Semester Will be Credited in Vth Semester.**
- **MOOCS introduced through SWAYAM in all semester.**



Program: Diploma
Semester: Three
Course: Mechanical Engineering
Course Code: 8D.201

L	T	P	C
3	0	0	3

Course Objective:

- To equip the mining engineering students with the basic principles of operation of Mining Machinery.
- Critically analyze problem and solve the problems related to mechanical elements and analyze the deformation behavior for different types of loads.
- Ability to understand fundamental material properties
- Ability to research and present on advance material topics
- To understand structure-properties relationship
- Manipulate atomic and micro-structural processes to create desired structure and properties
- To give basic knowledge of science behind materials and physical metallurgy
- To understand the stresses developed in bars, compounds bars, beams, shafts, cylinders and spheres.

Unit I

Units and Dimensions, Units of pressure, Volume, Temperature, Work, Power, Energy in SI system only and their dimensions. Engineering Materials: Chemical composition, properties and uses of following ferrous Metals: Cast iron, steel, Wrought iron, manganese steel, nickel steel, chromium steel, nickel-chromium steel, stainless steel. Non ferrous: Aluminum, copper, nickel, bronze, brass, copper nickel alloys, Aluminum alloys etc.

Unit II

Brakes and Clutches: Classification, Construction & working of block brakes, internal expanding brakes, hydraulic brakes, vacuum brakes (no numerical problems). Clutches: Construction & working of plate clutches, cone clutches, centrifugal clutch, and claw clutch (no numerical problems) IC Engine: Classification, Otto cycle, Diesel cycle. Two stroke & four stroke petrol engine .Two stroke & four stroke Diesel engine .Different systems like fuel injection, fuel ignition for petrol & diesel engines.

Unit III

Air Compressor: Classification, Definitions of different terms such as inlet pressure, discharge pressure, capacity, theoretical power, break power, free air delivery.

Unit IV

Hydraulics Machines: Properties of fluid, components of hydraulic circuits and their symbols, constructional details and working of hydraulic of shaper and hydraulic press. Types of pumps. Working principle of centrifugal pump, working principle of reciprocating pump. Uses of pumps in mining industry. Normalizing, annealing, hardening or quenching, tempering, case hardening of steel.

Suggested Reading:



1. *Strength of materials, Khurmi*
2. *Engineering Mechanics, Singer*
3. *Applied Mechanics, Ramanutham*

Program: Diploma

Semester: Three

Course: Mechanical Engineering Lab

Course Code: 8DP.201

L	T	P	C
0	0	2	1

List of Experiments:

1. Demonstration of different types of brakes.
2. Demonstration of different types of clutches and their working.
3. Demonstration of petrol engine (Two stroke & four stroke cycle engine.)
4. Demonstration of Diesel engine (four stroke cycle engine.)
5. Demonstration of reciprocating air compressor.
6. Demonstration & application of Rotary compressor.
7. Study of hydraulic circuit of shaper machine.
8. Study of hydraulic circuit of hydraulic press.
9. Demonstration of Centrifugal pump & its components.
10. Demonstration of Reciprocating pump & its components.



Program: Diploma
Semester: Three
Course: Mining Geology I
Course Code: 8D.202

L	T	P	C
3	0	0	3

Course Objective:

- This will provide basic knowledge on identification of various rocks & minerals by verifying the physical properties.
- The course will equip the students with the ability to appreciate the structure of earth and ore formations.
- The students will have knowledge about ore reserve estimation, ore assaying, remote sensing, geological mapping and identification of geological structures in the field.
- To practice the determination of engineering properties of rocks, preparation of weathering profiles, RMR, RQD, preparation of geological and structural maps and congestion of geological structures in the field.

Unit I

General geology: Branches, Sub branches, scope, origin of earth, continental Drift, Isostasy. Mineralogy: Definition, Classification of minerals and Physical and Chemical Properties of Minerals, properties of common minerals like Quartz, feldspar, Mica, Pyrite, Chalcopyrite, Galena, Hematite, Magnetite, Chromite, Psilomelane etc.

Unit II

Petrology: Igneous rocks: Magma and Lava, extrusive and intrusive forms. Classification and description of common igneous rocks (Granite, Dolerite, Gabbro, Basalt, Pegmatite)

Sedimentary Rocks: Sedimentation process; classification and description of common sedimentary rocks (Conglomerate, sandstone, shale, Limestone). Metamorphic Rocks: Process of metamorphism, classification & description of common metamorphic rocks (Slate, Marble, Quartzite, Gneiss, Schist)

Unit III

Paleontology and Stratigraphy: Concept of paleontology; fossils, their mode of presentation, concept of index fossils; principles of stratigraphic subdivisions and associated rock types of important are provenience & Coal belt.

Unit IV

Physical Geology: Erosion & weathering, River& wind erosion, Earth quake, Volcano

Geological Map: Drawing of Geological section of maps, Description of Geological maps, characteristics of contour line.

Suggested Reading:

1. A text book of Geology, P.K. Mukherjee



2. *Physical Geology*, A.K. Dutta
3. *Structural Geology*, S.W. Chiplonkar

Program: Diploma

Semester: Three

Course: Mining Geology I Lab

Course Code: 8DP.202

L	T	P	C
0	0	2	1

List of Experiments:

1. Study of physical properties of common rock forming Minerals and ore Minerals
2. Study of common igneous rocks.
3. Study of common sedimentary rocks.
4. Study of common metamorphic rocks with references to their structures, mineral composition and its uses.
5. Study and drawing of geological maps.



Program: Diploma
Semester: Three
Course: Mining Surveying I
Course Code: 8D.203

L	T	P	C
3	0	0	3

Course Objective:

- The course provides for basic skills in survey and correct usage of survey instruments.
- The student will be able to enter for the Elementary Survey Course.
- Appreciate the need for accurate and thorough note taking in field work to serve as a legal record
- Gain the ability to use modern survey equipment to measure angles and distances.
- Gain the ability to measure differences in elevation, draw and utilize contour plots, and calculate volumes for earthwork.
- Improve ability to function as a member of a survey party in completing the assigned field work.

Unit I

Introduction to surveying: Definition of surveying, objects of surveying, Plane and Geodetic surveying. Classification & Basic principles of surveying. Principle of chain surveying. Equipments in Chain surveying, cross staff, optical square its principle and use. Different operations in chain surveying, Ranging: direct & reciprocal ranging. Line ranger structure, principle of working and its use. Chaining: Chaining on flat & slopping ground, obstacle in chaining (No numerical). Errors in chaining. Offsetting.

Unit II

Compass Surveying: The Prismatic & Surveyors compass, their Comparison. Bearing of a line: Definitions: True & Magnetic Meridian; True and Magnetic bearings, Fore & Back bearings, Declination. Whole circle bearing system & Quadrantal. Bearing system. Conversion of bearings from one system to other. Calculation of angles from bearings. Calculation of bearings from angles. Local attraction: Sources, detection & its elimination. Magnetic Dip & Magnetic declination. Calculation of True bearings. Traversing with compass: Closed and open traverse; plotting a compass traverse; Checks for open & closed traverse; Closing error, Graphical adjustment of closing error. Introduction to Plane table and its accessories, Temporary adjustments of Plane table, centering, leveling, orienting the plane table by method of back sighting by method of magnetic needle. Methods of plane tabling Radiation, Intersection, Traversing, Resection method. Advantages & disadvantages of plane table survey, Errors in plane table survey.

Unit III

Levelling: Definitions of the terms used in Levelling. Concept of datum, Back sight, Foresight stations, change point, height of instrument. Dumpy and tilling level Construction and temporary adjustments. Levelling staff, their types. Balancing of back sight and Fore sight distances. Holding and Reading the staff, simple and differential levelling, and booking of readings. Reduction of levels by Collimation system and by Rise & fall system. Arithmetic check, computation of missing readings. Classification of leveling: Differential, Reciprocal, and fly levelling, Profile levelling, cross sectioning. Plotting of a profile and cross section. Difficulties in leveling, common mistakes in levelling. Permanent adjustments of Dumpy & Tilling level. Automatic level (General idea only). Transferring levels underground: Measurement of depth of vertical shaft. Establishing u/g bench mark in relation to surface bench mark. Marking center & grade line of drift, Marking Highest flood level. Study and use of Abney level, Auto set level, Temporary adjustments.



Unit IV

Contouring: Introduction and concept, definitions, purpose, Characteristic of Contour line, contour interval, factors affecting contour interval, Horizontal equivalent. Methods of Locating contours. Direct method, indirect method. Interpolation of contours by estimation, arithmetical and by graphical method. Plotting of contour maps. Uses of contour map. Measurement of areas & volumes by Simpson's and Trapezoidal, prizmoidal rule, measurement of stock of coal, overburden mines. Subsidence monitoring, data required for subsidence studies, parameters required during monitoring, layout of survey lines, survey stations, measurement techniques and Instrumentation, Measurement of displacement, slope and subsurface movement, frequency of subsidence measurement. General requirements of mine plan, Types of plan, copies of plans & sections to be submitted.

Suggested Reading:

1. *Surveying and leveling Vol. I & II* , T. P. Kanetkar
2. *Surveying & Levelling* , B.C. Punmia
3. *Surveying & Levelling* , Amarjit Aggarwal



Program: Diploma

Semester: Three

Course: Mining Surveying I Lab

Course Code: 8DP.203

L	T	P	C
0	0	2	1

List of Experiments:

1. Measurement of distance by Ranging and Chaining
2. Locating various object by chain & cross staff survey
3. Measurement of bearings of sides of traverse with prismatic compass and computation of correct included angle.
4. To find the difference in elevation between two points using Differential or Fly leveling.
5. Calculation of R.L. for different points involving 2 instrument stations & reduction by Height of Instrument & Rise and Fall methods.
6. To study different parts of a Transit Theodolite and Temporary Adjustments
7. To measure a horizontal angle by repetition method.

Program: Diploma

Semester: Three

L	T	P	C
3	0	0	3



Course: Introduction to Mining

Course Code: 8D.204

Course Objective:

- Enable students to demonstrate the importance of mining in national economy.
- Students will understand the terminology associated with the discipline and be familiar with the available regulatory mechanism to enable safe & sustainable mining operations.
- Equips the students with detail knowledge on various engineering techniques used for drilling, blasting, roof support and allied activities in mine construction for exploitation of minerals.

Unit I

Introduction to Mining & mineral: Definition, Different Types of Mining Methods, Mineral resources in India. Uses of important minerals mined e.g. Coal, lignite, iron ore, limestone copper, zinc, bauxite, gold, manganese, mica etc. Important Organizations involved like DGMS, CIL, MECL, CMRI, IBM etc. their role and functions.

Unit II

Mining Terminologies & definition: Common terminologies used in coal mining, metal mining, mine ventilation and environment and mine supports. Simple definition, explanation, purposes and sketches. Mode of access: Modes of entry to mine, Adit, Shaft, Trench, Open Pit and their description and sketches.

Unit III

Explosives & Accessories: Common explosive bases, Properties of Explosives, High Explosive & Low explosive, their comparison. Permitted explosives their types, composition, properties, uses, advantages & disadvantages. Brand names of some commonly used explosive of each type. Detonator, common types of detonators, plain detonators, instantaneous and delay action detonators their construction, uses, comparison etc. low tension & high-tension detonators. Safety fuses, detonating cords, detonating relays. Shot firing tools, exploders.

Unit IV

Introductory lessons on working Coal: Classifications of method of working coal (a) Board & Pillar (b) Open cast method (c) Long wall. Applicability condition for selection of each methods of working. Advantages, disadvantages & simple layout of each method.

Suggested Reading:

1. *Explosive and Blasting Techniques*, G.K. Pradhan
2. *Explosives and Blasting Techniques*, S.K. Das
3. *Element of Mining Technology Vol.- I*, D.J. Deshmukh



Program: Diploma
Semester: Three
Course: Applied Mathematics
Course Code: 9D.204

L	T	P	C
3	0	0	3

Course Objective:

- The subject helps the students to develop the fundamentals and basic concepts in vector calculus, ODE, Laplace transform and complex functions. Students will be able to solve problems related to engineering applications by using these techniques.
- Students will use mathematics concepts in real world situations.
- Students will simplify and perform operations with nonlinear expressions.
- Apply the principles of Vector algebra to solve a variety of basic problems in engineering and Applied Science.

Unit I

Complex Variable: Continuity, Differentiability, and analyticity of a function of a complex variable, Cauchy – Riemann equations in Cartesian and Polar form, Harmonic Function.

Unit II

Ordinary Differential Equation: Formation of Ordinary differential equation, Variable Separable Homogeneous Equation, Solution of Bernoulli's Equation, Exact-Differential Equation.

Unit III

Graph Theory: Graph, Sub graph, Walk travel and Path, Connected and Disconnected graph. Handshaking theorem or Edge and Vertex of a graph. Matrix representation of graph (Incidence and Adjacency Matrices). Spanning Tree. Kruskal's Algorithms for minimal Spanning tree.

Unit IV

Numerical Solution of Algebraic Equation and Simultaneous Equation: Bisection Method, Regula –Falsi Method, Newton-Raphson Method, and Gauss elimination Method, Iterative Method –Gauss Seidel and Jacobi's Method. Laplace Transform: Definition of Laplace Transform, Inverse Laplace Transform.

Suggested Readings:

1. *Higher Engineering Mathematics* –Dr.B.S.Grewal Khanna Publication
2. *Higher Engineering Mathematics* –H.K.Das
3. *Engineering Mathematic* -N.P.Bali Laxmi Publication
4. *Graph Theory: Prabhakar Gupta and Vineet Agarwal Pragati Prakashan.*



Program: Diploma

Semester: Three

Course: Communication and Soft Skills

Course Code: 40D.201

L	T	P	C
2	0	0	2

Course Objective: The aim is to develop students' soft skills, communication, leadership and teamwork skills; and personal development skills using practical approach and exposure of students to the realities of the world.

- To enhance Leadership – assessing the requirements of a task, identifying the strengths/weaknesses within the team, utilizing the diverse skills of the group to achieve the set *objectives*.
- To improve *Communication* – demonstrating clear briefing and listening /speaking skills.
- To make them realize that effective *communication* and interpersonal skills are crucial to increase employment opportunities and to compete successfully in the business environment.
- The *course* aims to cause a basic awareness about the significance of *soft skills* in professional and inter-personal *communications* and facilitate an all-round development of personality. Hard or technical skills help securing a basic position in one's life and career. But only *soft skills* can ensure a person retain it, climb.

Unit I

COMMUNICATION SKILLS

Introduction

Role of Communication in Today's World

Objective of Communication

Process of Communication

Elements of Communication

Essentials of Communication

Barriers/ Factors Inhibiting Communication

Flow of Communication

Verbal Mode of Communication

Unit II

COMMUNICATION NETWORK



Non Verbal Mode of Communication

Kinesics/Body Language, proxemics , chronemics, para lingual

Style in Technical Communication

Communication Skills; Reading, Writing, Speaking, Listening & Talking

Unit III

GRAPHICS

Introduction

Planning of Graphics

Placing of Graphics

Construction of Graphics

Types of Graphics (textual, visual, tables, bar Charts, pie charts, line charts, organizational charts, flow charts, maps & Pictographs)

Unit IV

TELEPHONIC CONVERSATION SKILLS

Introduction

Stages in Telephonic Conversation

Listening & Speaking Skills

Telephonic Skills

Problems in Telephonic Conversation

Intensive Listening

Suggested Readings:

- Monippally, Matthukutty. M. 2001. *Business Communication Strategies*. 11th Reprint. Tata McGraw-Hill. New Delhi
- Swets, Paul. W. 1983. *The Art of Talking So That People Will Listen: Getting Through to Family, Friends and Business Associates*. Prentice Hall Press. New York
- Lewis, Norman. 1991. *Word Power Made Easy*. Pocket Books
- Sen , Leena .*Communication Skills ; Eastern Economy Edition*



- Ghanekar , Dr. Anjali . *Essentials of Business Communication Skills* ; Everest Publishing House
- David Green . *Contemporary English Grammar, Structure & Composition* ; MacMillan
- Dictionary; Oxford
- Dictionary ; Longman

Websites

- www.tatamcgrawhill.com/digital_solutions/monippally
- www.dictionary.cambridge.org
- www.wordsmith.org
- www.edufind.com
- www.english_the_easy_eay.com

Program: Diploma
Semester: Three

L	T	P	C
3	0	0	0



Course: Disaster Management

Course Code: 14D.201

Course Objective:

- Develop an understanding of the key concepts, definitions and key perspectives of All Hazards Emergency Management
- Understand the Emergency/Disaster Management Cycle
- Have a basic understanding for the history of Emergency Management
- Develop a basic understanding of Prevention, Mitigation, Preparedness, Response and Recovery
- Develop a basic understanding for the role of public private partnerships.

Unit I

Understanding Disasters: Understanding the concepts and definitions of disaster, hazard, vulnerability, risk, importance, dimensions & scope of Disaster Management, Disaster Management cycle and disaster profile of India.

Unit II

Types, Trends, Causes, Consequences and Control of Disaster: Geological Disasters (earthquakes, landslides, tsunami, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunder-storms, hail storms, avalanches, droughts, cold and heat waves); Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear, bomb threat, explosion) and Man-made Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters; terrorist attack, , sudden shooting); Global Disaster Trends–Emerging Risks of Disasters–Climate Change and Urban Disasters; Financial emergency(risk of eviction, risk in arrears, sudden health emergency, family emergency, unexpected loss of income).

Unit III

Prevention and Mitigation of Disaster :Disaster Mitigation: meaning and concept, Disaster Mitigation Strategies Emerging Trends in Disaster Mitigation, Mitigation management, Role of Team and Coordination. Disaster Preparedness: Concept & Nature, Disaster Preparedness Plan, Preventions. Roles & Responsibilities of Different Agencies and Government, Technologies for Disaster Management. Early Warning System; Preparedness, Capacity Development; Awareness during Disaster.

Unit IV

Applications of Science and Technology for Disaster Management & Mitigation: Geo-informatics in Disaster Management (RS, GIS, GPS and RS) Disaster Communication System (Early Warning and Its Dissemination) Land Use Planning and Development, Regulations, Disaster Safe Designs and Constructions, Structural and Non Structural Mitigation of Disasters.

Suggested Reading:

1. *Disaster Management- J. P. Singhal, Laxmi Publications.*
2. *Disaster Management - Dr. Mrinalini Pandey, Wiley India Pvt. Ltd.*
3. *Disaster Science and Management- Tushar Bhattacharya, McGraw Hill Education (India) Pvt. Ltd.*



4. *Disaster Management: Future Challenges and Opportunities* - Jagbir Singh, K W Publishers Pvt. Ltd.

(For Lateral Students only)

Program: Diploma
Semester: Three

L	T	P	C
2	0	0	0



Course: Life Skills I

Course Code: 40D.101

Course Objective:

To impart basic skills of Professional Communication in English through intensive practice to the Students, so as to enable them to function confidently & effectively in that Language in the Professional Sphere of their life

- The student must have some basic command of English so that the Student must be able to:
- Write reasonably & grammatically
- Understand (if not use) at least some 2500 general purpose words of English
- Use some 2000 (at least 1500) general-purpose words of English to express himself/herself in writing & 1500 such words to talk about day-to-day events & experiences of life.
- Understand slowly-delivered spoken material in Standard Indian English, and
- Speak reasonably clearly (if not fluently) on routine matters with his fellow Students, with proper word stress, intonation pattern, accent and perfect articulation

Unit I

Basic Grammar

- Noun, Verb, Adverb, Adjective & Preposition
- Sentence
- Tense: Present ,Past & Future
- Voice
- Narration
- Concord
- English Modals
- Connectives
- Degree of Comparison
- Nominalization

Unit II

Practice Exercise

- Re-Writing Sentences
- Gap Filling
- Common Errors
- Phrases & Idioms
- Homophones (Commonly Confused Words)
- Vocabulary Building
- Word Quiz

Unit III

Written Communication Skills

- Requisites of good sentence writing
- Effective sentence structure
- Sentence Building/ Sentence coherence



- Sentence Emphasis/theme
- Development of a paragraph
- Paragraph structure
- Principles of paragraph Writing
- Paragraph length/ coherence/ Division

Unit IV

Etiquettes & Manners

- Dining etiquettes
- Workplace etiquettes
- Professional Manners
- Social Etiquettes
- Group Behavior
- Tour & Travel Etiquettes

Suggested Readings:

1. Monippally, Matthukutty. M. 2001. *Business Communication Strategies*. 11th Reprint. Tata McGraw-Hill. New Delhi
2. Swets, Paul. W. 1983. *The Art of Talking So That People Will Listen: Getting Through to Family, Friends and Business Associates*. Prentice Hall Press. New York
3. Lewis, Norman. 1991. *Word Power Made Easy*. Pocket Books
4. Sen, Leena. *Communication Skills*; Eastern Economy Edition
5. Ghanekar, Dr. Anjali. *Essentials of Business Communication Skills*; Everest Publishing House
6. David Green. *Contemporary English Grammar, Structure & Composition*; MacMillan
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- www.english_grammar_lessons.com
- www.wikipedia.org/wiki/english_grammar



DIPLOMA IN MINING ENGINEERING												
SEMESTER IV												
S. No	Subject code	Name of Subject	Period			Evaluation Scheme				Subject	Credit	Hours
			L	T	P	Assignment	T A	Total	ESC			
1	8D.251	Mining Machinery I	3	0	0	20	10	30	70	100	3	3
2	8D.252	Basic Mine Electrical Engineering	3	0	0	20	10	30	70	100	3	3
3	8D.253	Mining Geology II	3	0	0	20	10	30	70	100	3	3
4	8D.254	Underground Coal Mining	3	0	0	20	10	30	70	100	3	3
5	8D.255	Underground Metal Mining	3	0	0	20	10	30	70	100	3	3
6	40D.251	Professional Skills	2	0	0	20	10	30	70	100	2	2
7	40D.151	***Life Skills II	2	0	0	20	10	30	70	100	0	2
PRACTICAL/ SESSIONAL												
1	8DP.252	Basic Mine Electrical Engineering Lab	0	0	2	0	30	30	20	50	1	2
2	8DP.253	Mining Geology II Lab	0	0	2	0	30	30	20	50	1	2
								Total		700	19	21

- Note: Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of IVth Semester Will be Credited in VIth Semester.
- MOOCS introduced through SWAYAM in all semester.
- ** NOTE: Qualifying Non Credit Course & only Viva voce is conducted



Program: Diploma

Semester: Four

Course: Mining Machinery I

Course Code: 8D.251

L	T	P	C
3	0	0	3

Course Objective:

- Enables the students to be familiar with the various machines used in opencast mine, their construction and performance.
- The students will get exposure towards the material handling methods and systems and its principle to convey the minerals or materials from mines, plants and workshops.
- Enables the students to select appropriate machinery for various mining operations based on the production targets.
- Enables students about the conveyor system and its advancement.

Unit I

Transport of ore: Different types of haulages, Description of each haulage system, Direct rope haulage, Endless rope haulage, Main and Tail rope haulage, Gravity haulage Calculation of rope haulage, Calculation of TTF, Power of the rope, Safety devices used on rope haulage system Stop block/Buffers, Back stay, Monkey catch, Age craft Device, Runaway Switch, Drop Warwick. Locomotive haulage: Different types/ Applicability, Diesel locomotive, Electric locomotive, Air compressed locomotive, Battery locomotives. Definition of draw bar pull, Ideal gradient, super elevation. Different Types of Conveyor, Chain conveyor, Plate conveyor, Belt conveyor, Condition of Suitability of each type, Advantageous and disadvantages.

Unit II

Winding shaft: Purpose of Winding, Main equipments used for Winding, Head gear, Headgear pulley, Cage/Skip, Winding Rope, Winding drum, Guides, Keps, Suspension Gear, Electric motor. Different types of winding, Drum winding, Koepe Winding, Drum winding different types, Cylindrical drum, Conical drum, Cylindro conical drum, Bi-cylindro conical drum, Provisions on winding drum, Flanges, Depth indicator, Mechanical Brakes (different types), Automatic Contrivance, Angle of fleet, Guides different types, Rigid guides, Flexible cable, Causes of cage oscillation, Cage suspension Gear Rope Capel, D link and bull chain, Safety hook, Triangular distribution plate, Bridle chain, Different types of kep, Rigid keps, Davies improved keps gear, List of safety devices used in winding, Characteristics curves, Koepe winding.

Unit III

Wire ropes: Classification of different types of wire ropes, Stranded rope and Non stranded rope Different types of stranded rope, Different types of Non stranded rope, Lays of rope, Different definition like Space factor, static load, dynamic load, factor of safety.

Unit IV

Selection of wire rope, Care and maintenance in ropes, Types of deterioration in the ropes, testing of wire ropes, Types of Rope capping, White metal capping (cone socket type capel), Wedge type capping (Reliance rope capel), Capping with split capel and rivets (Split capel), Recapping and Rope splicing procedure.



Suggested Reading:

1. *D.J. Deshmukh ,VOL- III*
2. *Mine pump, haulage, winding, S. Ghatak*



Program: Diploma

Semester: Four

Course: Basic Mine Electrical Engineering

Course Code: 8D.252

L	T	P	C
3	0	0	3

Course Objective:

- To understand the electrical layouts and power distribution in mine.
- To study the various modes of transport means and electrical circuits.
- To study the types of electrical equipment used in mines.
- Students should be able to verify the principles studied in theory by performing experiments in the laboratory.

Unit I

Electric circuit: Resistance, Current, Voltage, Work, Power and Energy Ohm's Law, AC Current Three phase & Single phase, Storage Batteries- Constructing & working

Unit II

DC Machine: Construction & principles of operating, Magnetization and load characteristics of series shunt and compound generators and motors. Motor starter, speed control and their field of applications. Construction and principles of operation, Different types of A.C motors, methods of starting and speed control of Induction motor, Universal motor.

Unit III

Power supply system: Transmission & distributing of Electrical power by overhead lines and cables Types of cables, layout of underground cables, shaft cables protection system and switchgear like Relays, circuit breaker and fuses. Earthing and types of earthing.

Unit IV

Single phase transformers: Construction and principles of operation, types of transformers, Efficiency and Regulations, Auto transformer.

Suggested Reading:

1. *Electrical Technology*, A.K. Theraja
2. *Electrical Engineering In mines*, N.K Dutta



Program: Diploma

Semester: Four

Course: Mine Electrical Engineering Lab

Course Code: 8DP.252

L	T	P	C
0	0	2	1

List of Experiments:

1. Measurement of Resistance by voltmeter, Ammeter method.
2. Magnetization curve of DC machine.
3. Load characteristics of DC shunt generator
4. Load characteristics of DC series generator.
5. Speed control of DC shunt motor.
6. Speed-load Characteristics of Induction motor
7. Voltage and current ratios of transformer
8. Determination of efficiency and regulating of transformer.
9. Study of DC shunt motor starter to understand normal principles of working.
10. Study of Induction motor starters to understand normal principles of working.



Program: Diploma
Semester: Four
Course: Mining Geology II
Course Code: 8D.253

L	T	P	C
3	0	0	3

Course Objective:

- The course provides for basic knowledge on economics of ore, exploration and practical site investigation
- The students will have familiarity about economic geology and Indian mineral deposits. They will have deep knowledge about geophysics, remote sensing and GIS.
- The students will have knowledge about ore reserve estimation, ore assaying, remote sensing, geological mapping and identification of geological structures in the field.

Unit I

Structural Geology :Study of Topographic maps, unconformities, folds, faults, and joints-their nomenclature , classification and recognition, forms of igneous intrusion- Dyke, sill and batholiths, effect of fold & fractures on strata pre bodies.

Unit II

Economic geology: Ore and Gangue, process of Ore formation, major Indian mineral deposits (Iron, Manganese, Copper, Lead, Zinc) - distribution & mode of occurrence, resources and reserves. Exploration & prospecting: Different methods of prospecting, Geological – Geophysical & Geochemical, Geographical information system & remote sensing.

Unit III

Coal Geology :Rank, characteristics and important constituents of coal, classification and origin of Coal chief characteristics of Indian Coals, Geology of principle Coalfields of India Stratigraphy: Principles of Stratigraphy, Geological Time-scale, Physiographic Divisions of India, Major geological formations of India, Archean & Dharwar System with Economic importance, Cuddapah system with economic importance, Vindhyan System with economic importance, Gondwana super group with economic importance, Deccan traps.

Unit IV

Petroleum geology: Properties & Uses of Petroleum, Origin of Petroleum and Occurrence of Petroleum in India.

**Suggested Reading:**

1. *A text book of Geology*, P.K. Mukherjee
2. *Physical Geology*, A.K. Dutta

Program: Diploma

Semester: Four

Course: Mining Geology II Lab

Course Code: 8DP.253

L	T	P	C
0	0	2	1

List of Experiments:

1. Study of Topographical maps.
2. Calculation of attitude and various methods.
3. Thickness and depth of Ore bodies.
4. Maps illustrating fold, fault and unconformity.
5. Crystal system of common minerals.



Program: Diploma
Semester: Four
Course: Underground Coal Mining
Course Code: 8D.254

L	T	P	C
3	0	0	3

Course Objective:

- Enables the students to appreciate various coal mining methods and to ably manage highly mechanized mines.
- The students will have good exposure about the various advanced methods of coal
- Mining with the knowledge about advanced coal face mechanization.
- To introduce the recent trends of level of mechanization for coal face.
- To understand the various advanced methods of coal mining.

Unit I

Mine Supports: Steel support, Steel arches, Yielding supports, Friction props, and hydraulic props, description, applicability and comparison, Powered support, Fore poling, Long wall face support, Withdrawal of support from goaf.

Unit II

Working Coal: Board and Pillar Method, classification of method of working, Conditions for selection of method of working, Board and pillar applicability advantages and disadvantages, Calculation of percentage of extraction, Design of panel, Different layout classification, SDL chain conveyor layout, Continuous miners, LHD layout, Manpower calculation and OMS, Open & closed panel system, Preparatory arrangements before depillaring, Line of extraction and numbering of pillars, systematic support rules, Different types of pillar extraction methods Caving: Stowing, Thick seam working, Precautions while working near restricted areas, Working near fire area, Working below waterlogged area, Working below depillared goaf, Contiguous working of seams, Layout and case study

Unit III

Longwall mining: Applicability, Design of long wall panel, Factors affecting length of long wall face, Barrier width, gate road length, Long wall advancing, Long wall retreating, Cyclic long wall and Non cyclic long



wall, Different machines used, Layout of DERD manpower calculation, Thick seam working with long wall. Top slicing and sub level caving, long wall caving and stowing.

Unit IV

Subsidence in mining: Theories of subsidence, Different definition related to subsidence, Different factors affecting subsidence, precautions to be taken on surface in subsidence area, measurement of subsidence and preservation of record.

Suggested Reading:

1. *Underground Winning of Coal*, T.N.Singh
2. *Modern Coal Mining Technology*, S.K. Das

Program: Diploma

L	T	P	C
3	0	0	3



Semester: Four

Course: Underground Metal Mining

Course Code: 8D.255

Course Objective:

- Enables the students to select suitable methods of working underground metal mines and decide the necessary parameters of mine construction.
- The students will have good knowledge about the various advanced methods of metal Mining and special mining techniques to overcome the field issues.
- To understand the various advanced and recent methods of metal mining.
- The students will have basic concept on metal mining methods, mine design, development and operations of metal mines. They will also know about novel methods of metal mining and its applications.

Unit I

Underground development: Terminology used in metal mines, Types of ore bodies, Types of ore bodies, Types of underground opening, location of openings, Opening up of a mineral deposit by vertical shaft, inclined shaft, Adit, Level interval, factor considered while deciding level interval/length of back, Drivages of raises and winzes, Driving manually, Modern methods Alimak, long hole method, Drop raising, raise boring, Common supports in metal mines.

Unit II

Unsupported Stoping Methods: Classification and choice of stoping, Methods, Open stoping methods, underhand, overhand, Breast stoping, sublevel stoping, Blast hole stoping, VCR, Shrinkage method, their conditions of Applicability, Sequence of Development, stoping operation, Cycle of operations etc. System of removal of ore from stope. Supported & caving methods: Artificially supported methods cut and fill, square set, stull stoping Methods. Their applicability, stope Preparation, stoping operation, cycle of operation, relative merits and demerits etc. Caving methods - Top slicing, Sublevel caving and block caving methods, applicability, stope preparation stoping, cycle of operation etc.

Unit III

Shaft Sinking: Size, shape, Factors considered for location of shaft, marking center, and shaft-centering arrangement, sinking of shaft below rock head- operation of drilling, charging and blasting and mucking operation. Disposal of debris, Shaft lining: Temporary lining, Permanent lining of shaft: Brick, monolithic, reinforced concrete lining, shaft tubing's etc, Walling scaffold, rider, ledge formation, underpinning, water garland crib etc. Special method of shaft sinking: Different special methods of shaft sinking, condition of applicability of each method, Description etc, Widening and Deepening of shaft.

Unit IV

Boring: Purpose of boring, classification of boring methods, applicability of boring methods, Drill Bits for various types of drilling/boring, Surface arrangement, assembly, working of Rotary boring, Screw and hydraulic feed mechanism, Core recovery, core barrels, Recover of broken tools, Bits, Bore hole survey, Deviation of boreholes.

Suggested Reading:

1. *Elements of Mining Technology Vol I, II, D.J. Deshmukh*



Program: Diploma
Semester: Four
Course: Professional Skills
Course Code: 40D.251

L	T	P	C
2	0	0	2

Course Objective: It is student-centric, value based, activity oriented professional education, where the Faculty is not only the disseminator of common wealth of knowledge and experience but the organizer of learning situations, facilitator of the learning process and co coordinator of learning following the age old adage of "I hear, I forget, I see, I remember, I do, I understand"

- In this unit the students get opportunities to apply their classroom learning to practical situation.
- This course aims to develop the professional traits in them, so that they can meet the neo-challenges of job opportunities.
- Students become the architect of their career goals.
- Acquire leadership traits,
- Interpersonal skills,
- Adaptability, discussion skills, interview skills etc.

Unit I

DISCUSSION SKILLS

Introduction
Importance of Group Discussion Skills
Process, Scope & Limits of Group Discussion
Group Discussion, Interaction Strategies, Individual Contribution
Leadership Skills, Team Management, Creating Friendly Co-operative Atmosphere
Selection Group Discussion, Interactive Oral Process, Purposeful & Goal Oriented
Characteristics, Agreement on Group Goals, Agreement on Procedure, Effective
Communication, Equitable Distribution of Time; Speaking & Listening Skills; Adaptability;
Assertiveness; Command Over the Subject

Unit II

NEGOTIATION SKILLS

Speaking & Listening Skills
Rapport Building
Decision Making Ability
Problem Solving Skill
Attitudes
Adaptability
Conflict Handling Ability

Unit III

JOB SEARCH & CORRESPONDENCE SKILLS

Introduction; Job Search Strategies



Developing Job Communication Skills
Skill Analysis
Job Communication Process
Creating Network,
Prelude; Biodata, Curricula Vitae (CV) Resume
Determining the Need of the Employer

Unit IV

INTERVIEW SKILLS

Interview; Introduction
The Interview Process
Types of Interview; Face to Face, Group Interview, Through Video Conferencing, Telephonic, Skype, Panel Interview
Planning/Purpose
Pre-Interview Techniques
Answering Strategies
Follow up

Suggested Readings:

1. Monippally, Matthukutty. M. 2001. *Business Communication Strategies*. 11th Reprint. Tata McGraw-Hill. New Delhi
2. Swets, Paul. W. 1983. *The Art of Talking So That People Will Listen: Getting Through to Family, Friends and Business Associates*. Prentice Hall Press. New York
3. Lewis, Norman. 1991. *Word Power Made Easy*. Pocket Books
4. Sen , Leena .*Communication Skills ; Eastern Economy Edition*
5. Ghanekar, Dr. Anjali. *Essentials of Business Communication Skills ; Everest Publishing House*
6. David Green. *Contemporary English Grammar, Structure & Composition ; MacMillan*
7. *Dictionary; Oxford*
8. *Dictionary ; Longman*

Websites

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- www.dictionary.cambridge.org
- www.wordsmith.org
- www.edufind.com
- www.english_the_easy_eay.com
- www.englishclub.com
- www.english_grammar_lessons.com



(For Lateral Students only)

Program: Diploma

Semester: Second

Course: Life Skills II

Course Code: 40D.151

L	T	P	C
2	0	0	0

Course Objective:

To impart basic skills of Professional Communication in English through intensive practice to the Students, so as to enable them to function confidently & effectively in that Language in the Professional Sphere of their life.

The student must have some basic command of English so that the Student must be able to:

- At the end of the course the student should become a good communicator not only in the organization but in day today life also. Should know and learn the dynamics of external and internal communication.
- Use some 2000 (at least 1500) general-purpose words of English to express himself/herself in writing & 1500 such words to talk about day-to-day events & experiences of life.
- Understand slowly-delivered spoken material in Standard Indian English, and
- Speak reasonably clearly (if not fluently) on routine matters with his fellow Students, with proper word stress, intonation pattern, accent and perfect articulation.
- Should have command over the language.

WRITING SKILLS

Unit I

Letter Writing

- Business/official Letters
- Letter Writing Skills
- Planning of the Letter
- Letter Writing Process
- Form & Structure
- Essentials of Letter Writing
- Types of Professional Letters: letter of enquiry, letter of placing order, information seeking letter, letter of claim & complaint, information giving letter, letter of acceptance, letter of rejection

Unit II

Professional Writing

- Job Application, introduction, layout & format (specimen)
- D O letter
- Resume & Job Application
- Covering Letter
- Editorial Letter



- Writing Mails & SMS (E-Language)
- Notice, Memo, Circular & Minutes Writing.
- Social Letters (letters to friends/relatives etc.)

STUDY SKILLS

Unit III

Reading Skills

- Newspaper Reading
- Mechanics of Note making
- Note Making Techniques/ Reduction Devices
- Organization Techniques/Method of Sequencing
- Mechanics of Summarizing
- Outlining & Paraphrasing

Unit IV

Referencing Skills

- Referencing Skills
- Method of Referencing
- Using Foot Notes
- Scanning and Skimming Skills
- Finding required Information/Meaning/ Pronunciation

Suggested Readings:

1. Monippally, Matthukutty. M. 2001. *Business Communication Strategies*. 11th Reprint. Tata McGraw-Hill. New Delhi
2. Swets, Paul. W. 1983. *The Art of Talking So That People Will Listen: Getting Through to Family, Friends and Business Associates*. Prentice Hall Press. New York
3. Lewis, Norman. 1991. *Word Power Made Easy*. Pocket Books
4. Sen, Leena. *Communication Skills ; Eastern Economy Edition*
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 - www.englishclub.com
 - www.english_grammar_lessons.com
 - www.wikipedia.org/wiki/english_grammar
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DIPLOMA IN MINING ENGINEERING												
SEMESTER V												
			Period			Evaluation Scheme				Subject	Credit	Hours
S. No	Subject code	Name of Subject	L	T	P	Assignment	T A	Total	ESC			
1	8D.301	Surface Mining	3	0	0	20	10	30	70	100	3	3
2	8D.302	Mine Surveying II	3	0	0	20	10	30	70	100	3	3
3	8D.303	Mining Machinery II	3	0	0	20	10	30	70	100	3	3
4	8D.304	Mine Ventilation	3	0	0	20	10	30	70	100	3	3
5	ELECTIVE I (Any one of the followings)											
	8D.331	Mine Environmental Engineering	3	0	0	20	10	30	70	100	3	3
	8D.332	Mine Reclamation and Closure										
	8D.333	Mine Planning and Design										
6	40D.401	**Seminar in Executive Communication	2	0	0	20	10	30	20	50	0	2
PRACTICAL / SESSIONAL												
1	8DP.302	Mine Surveying II Lab	0	0	2	0	30	30	20	50	1	2
2	8DP.304	Mine ventilation & Environmental Engineering Lab	0	0	2	0	30	30	20	50	1	2
3	8D.345	Practical Training Project I	0	0	0	-	-	-	-	50	2	2
									Total	650	19	21

- **** NOTE: Qualifying Non Credit Course & only Viva voce is conducted.**
- **MOOCS introduced through SWAYAM in all semester.**



Program: Diploma
Semester: Five
Course: Surface Mining
Course Code: 8D.301

L	T	P	C
3	0	0	3

Course Objective:

- Enables the students to apply the knowledge of mining machinery in their economic deployment for achieving the production targets in opencast mines.
- To introduce the various techniques for mine planning, geotechnical investigation and equipment management.
- To appreciate the modern trends in opencast mines, safety and environment.
- The students will have insight about the advanced techniques for mine planning, geotechnical investigation and equipment management and also will understand the modern trends in opencast mines, safety and environment.
- The students will have ability to classify and select the suitable surface mining methods and equipment based on site conditions. They will also have a concept of waste dump formations and slope failures in surface mines.

Unit I

Introduction to opencast mining:

Classification of surface mining methods, classification based on mechanization .Factors affecting choice of opencast mining methods, Stripping ratio: Maximum allowable stripping ratio, overall stripping ratio, Break even stripping ratio. Advantages and disadvantages of opencast mining. Elements of Benches: Height, width, angle of slope, toe, crest, statutory provisions regarding height, width, angle of slope etc. Opening up of deposit:Unit operations involved, site preparation, Box cut, entry system in opencast mines. Opencast mine layout, factor determining choices of layout, overburden excavation, Disposal of overburden, overcasting etc, sample layouts for lime stone, copper, coal , iron ore deposits, method of work, machines required , manpower, OMS etc.

Unit II

Opencast mining machinery: Classification of Excavating

equipments, selection, choices of opencast mining machinery, Excavators shovel, Rope shovel, hydraulic shovel, application, advantages, disadvantages, comparison Rope shovel and hydraulic shovel, operating parameter, output of a shovel. Various attachments to shovel, Specifications. Back hoe, operating parameter, application. Dragline, operating parameters, applicability, working, advantages, disadvantages, comparison with shovel. Specifications, Bucket wheel and Bucket chain excavators, Application, advantages & disadvantages, operation, working methods by Bucket wheel excavator, Rippers, Scrappers, bulldozer etc, Surface miner its application, working.

Unit III

Opencast explosives: Explosives used in opencast mine, AN/FO, slurry explosive, emulsion explosives, Heavy AN/FO explosive, LOX, their properties, composition etc. Boosters. Initiation system, non electric



initiation system, Raydets, Nonel, shock tubes, electronic detonators, etc. Bulk explosive system, site mixed slurry, site mixed Emulsion, Bulk-loading system. Advantages, comparison. ANFO precautions while mixing, handling and use, Conditions for using bulk explosives.

Unit IV

Blasting practice in opencast mines: Bench blasting terminology, Blast hole geometry, hole depth, burden, spacing, sub grade drilling, bottom charge, column charge, stemming height. Factors to be considered while blast designing. Simple numerical on blast design for the bench of surfaces mine. Single and multiple rows blasting their comparison, Sequence of blasting in single & multiple rows. Precautions while charging and firing of holes in deep hole blasting, deck charging, muffled blasting, control blasting techniques, secondary blasting/breaking in opencast mines, transport of Explosives in bulk.

Suggested Reading:

1. *Explosive and Blasting Techniques*, G.K. Pradhan
2. *Surface Mining Technology*, S.K. Das
3. *Elements of Mining Technology Vol I*, D.J. Deshmukh



E

Program: Diploma
Semester: Five
Course: Mine Surveying II
Course Code: 8D.302

L	T	P	C
3	0	0	3

Course Objective:

- The student will enable to carry out underground surveys and stope measuring.
- Plotting contour plans
- Understand advanced surveying techniques such as remote sensing and geodetic surveying.
- Enables the students to select and use suitable survey instruments for data needed in planning and executing the mineral extraction.

Unit I

Theodolite: Classification of theodolite, definitions and terms used in operating theodolite, Temporary adjustments of transit theodolite. Fundamental axes of theodolite, Measurement of horizontal and vertical angles, Observe magnetic bearing of a line, Use of theodolite for Prolonging a straight line, for lining in, Lay-off horizontal angle, Sources of errors in theodolite work and their elimination. Permissible errors in mine surveying. Theodolite traversing: Methods of traversing, by included angles & Deflection angles, Checks in closed & open traverse, Traverse computation: latitude, departure, Consecutive coordinates, independent coordinates, error of closure. Adjustment of closed traverse, balancing by Bowditch and transit rule, Gales Traverse table, Permanent adjustments of theodolite. (Only relationship of different axis of theodolite)

Unit II

Geodetic Triangulation: Basic principles, Triangulation figures, Classification, Base line measurement. Correction to Base line Measurement. (Elementary idea only). Tachometric surveying: Instruments, Requirement of a tachometer. Tachometric methods: Fixed hair method; Reduction formula for horizontal distance & elevation with horizontal sights. Setting out of curves: Classification of curves. Definition of various parts of curves, Elements of simple circular curve, Relation between radius of curve & degree of curve. Setting out curves by linear methods and single theodolite method, super elevation or cant. Purpose of super elevation, Study and use of Planimeter including digital planimeter.

Unit III



Underground surveying: Marking of underground stations, Correlation Survey- Direct traversing through incline, by plumbing wires in shaft, co planning method, Weisbach's triangle method, Weiss quadrilateral method, Gyro- theodolite method. Stope survey: Definition & Purpose of stope survey, Classification of stope surveying, Methods of stope surveying Stop survey flat deposit: Tape triangulation, Tie Method, instrumental method Stope survey in steeply inclined deposit. Stop survey in massive ore bodies, irregular in dip & shape.

Unit IV

Advances in mine surveying: Global Positioning System; Principle, The system, application in Opencast mine surveying, Advantages and Disadvantages. Total station; application in distance measurement, angle measurement, Electronic Distance Measurement; Principle of measurement, types etc. Gyro theodolite; principle of gyro, gyro attachments and Laser plummet. Dip & fault: Introduction, Definition of Dip, Strike Apparent dip, Full dip, Numerical for determining true and apparent dip, stratigraphically thickness of seam. Definition of Fault. Normal fault, Reverse fault. Numerical on fault problems, connection of mine plan to national Grid.

Suggested Reading:

1. *Surveying and leveling* , T. P. Kanetkar
2. *Surveying & Levelling* , B.C. Punmia

Program: Diploma
Semester: Five

L	T	P	C
0	0	2	1



Course: Mine Surveying II Lab

Course Code: 8DP.302

List of Experiments:

1. Demonstration of theodolite reading the vernier.
2. Temporary adjustments of theodolite & measurement of horizontal angle by repetition method.
3. Measurement of horizontal angle by reiteration method.
4. Observation of magnetic bearing of a line by Theodolite by compass attachment.
5. Measurement of deflection angle by Theodolite by taking an open traverse of 4 – 5 sides.
6. Prolongation of a survey line with the help of a Theodolite.
7. Measurement of Vertical angles using theodolite.
8. To Find constants of a given tacheometer.
9. Determining the reduced level and horizontal distance of an inaccessible object by Tachometer.
10. Setting out a simple circular curve by offsets from long chord
11. Setting out a simple circular curve by sub cord produced method.
12. Setting out a simple circular curve by (single Theodolite method) Rankine's method.
13. Demonstration of EDM.
14. Demonstration of micro optic theodolite.
15. Demonstration of Total station.

Program: Diploma



Semester: Five
Course: Mining Machinery II
Course Code: 8D.303

L	T	P	C
3	0	0	3

Course Objective:

- Enables the students to select appropriate machinery for various mining operations based on the production targets.
- The students will get exposure towards the material handling methods and systems and its principle to convey the minerals or materials from mines, plants and workshops.
- The students will have practical knowledge about underground machineries, ropes, and pit-top and pit-bottom layouts.
- Enables students about the conveyor system and its advancement.
- The students will have basic knowledge on motive power used in mines, pumping, rope haulage and other transport systems. They also will know about mine electrical engineering in all statutory aspects.

Unit I

Mine pumps: Sources of water in Mines, Classification of Mine Pumps, Reciprocating Pump, Single acting, double acting and Ram pumps. Centrifugal Pumps, Turbine Pumps, Installation of pump, Operation of pump, Fitting on pump, Starting and stopping of pump, Face pumps, Characteristics Curves of Centrifugal and turbine pumps, Calculations for pump discharge etc, Snoring of pump, its prevention, Water hammer.

Unit II

Coal Cutting Machine: Purpose of coal cutting machine, Classification of coal cutting. Machine, Different types of cut, Different parts of CCM, Introduction of continuous miner, Different types of mechanical Loaders. Drills for Coal and Stone: Various types, Jumbo drills.

Unit III

Electric Power Supply: Types of cables used in mines Permanent cable, Different types, construction Semi flexible cable, Different types, construction Flexible cable, Different types, construction. Cable care and maintenance during Use and storage.

Unit IV

Prime Mover for Mining Machinery: elements of mechanical power transmission -gear, belt, chain, coupling, clutch and brake.

Suggested Reading:

1. D.J. DESHMUKH VOL I, II
2. Mine pump, haulage, winding, S. GHATAK



Program: Diploma
Semester: Five
Course: Mine Ventilation
Course Code: 8D.304

L	T	P	C
3	0	0	3

Course Objective:

- Enables the students to understand the airflow in sufficient quantity and quality to dilute contaminants to safe concentrations in all parts of the facility where personnel are required to work or travel.
- Enables the students to understand the ventilation requirements.
- Students will be able in selecting appropriate mine fans and design suitable ventilation structures.
- To learn ventilation methods and strata monitoring instruments.
- The students will have better understanding about methods of ventilating longwall faces and Bord and pillar method of mining.

Unit I

Mine Air: Different Gases / Damps found in mines, Definition of damps, their threshold limits, physiological effects, source of production and detection, Degree of gassiness of seam. Flame safety lamps, its principle, construction, safety features, and comparison. Detection of Methane by flame safety lamp, MSA Methanometer its principle of working, construction. Principle of other method of detection of methane (description of equipment not required)

Unit II

Mine Climate: Purpose and standards of ventilation, standards for minimum & maximum velocity of air for different locations, Pressure, ventilating press, water gauge, Temperature, sources of heat in mines, Moisture content of mine air relative humidity, wet bulb temperature, measurement of relative humidity, Cooling power of mine air, determination of cooling power, methods of improving cooling power of mine air, effect of heat and humidity on miners.

Unit III

Natural Ventilation: Natural ventilation Pressure, geothermic gradient, Factors causing NVP, Effect of seasonal changes on direction of Natural ventilation, limitation of Natural ventilation. Motive column, calculation of natural ventilation pressure. Artificial Ventilation: *Different types of fans used in mines: centrifugal & axial flow, their principle of working, Exhaust & forcing type. Purposes of evasee & volute casing. Reversal of air current, and characteristics curves of fans. Fans in series and parallel, Comparison between axial flow & Centrifugal fan; exhaust & forcing Fan. Fan laws, Manometric efficiency overall efficiency, theoretical depression produced by fan. Numerical on fan laws.*

Unit IV

Distributions & coursing of Air in Mines: Laws of air flow in Mines, Atkinson's formula splitting, advantages & disadvantages, Numerical on splitting, equivalent orifice. Numerical on equivalent orifice. Ventilation appliances, Auxiliary ventilation: Different methods, advantages & disadvantages, hazards associated with auxiliary ventilation, precautions required .Booster fan: purpose, dangers associated, Precautions before installation. Numerical on Booster fan, Accessional



and Descensional ventilation, Advantages and disadvantages. Ventilation Survey: scope and importance of ventilation survey, survey interval and location of survey station, ventilation plan. Measurement of quantity & pressure difference, anemometer, pitot static tube, Manometer, quantity survey.

Suggested Reading:

1. *D.J. DESHMUKH VOL I, II*
2. *Mine Environment & Ventilation, G.B. Mishra*
3. *Mine Disaster & Mine Rescue, M.A. Ramlu*



Program: Diploma

Semester: Five

Course: Mine Ventilation and Environmental Engg. Lab

Course Code: 8DP.304

L	T	P	C
0	0	2	1

List of Experiments:

1. Demonstration of CO-detector and measurement of carbon monoxide using CO-detector.
2. Demonstration of MSA Methanometer and measurement of methane using Methanometer.
3. Dismantling & assembling of different types of Flame safety lamps.
4. Detection of Methane using flame safety lamp.
5. Demonstration of whirling hygrometer and determination of relative humidity using whirling hygrometer.
6. Demonstration of Kata thermometer and determination of cooling power by Kata thermometer.
7. Demonstration of various ventilation devices.
8. Demonstration of vane Anemometer and determination of quantity by Anemometer.
9. Demonstration of velometer and measurement of air velocity by velometer.

ELECTIVE -I

Program: Diploma



Semester: Five

Course: Mine Environmental Engineering

Course Code: 8D.331

L	T	P	C
3	0	0	3

Course Objective:

- Enables the students to understand the airflow in sufficient quantity and quality to dilute contaminants to safe concentrations in all parts of the facility where personnel are required to work or travel.
- Enables the students to understand the ventilation requirements.
- Students will be able in selecting appropriate mine fans and design suitable ventilation structures.
- To learn ventilation methods and strata monitoring instruments.
- The students will have better understanding about methods of ventilating longwall faces and Bord and pillar method of mining.

Unit I

Mine Fires: Surface fire and underground fires, their causes, detection and prevention. Classification of fires, Portable fire extinguishers their applicability, places of portable fire extinguishers in mines, maintenance of fire extinguishers. Spontaneous heating: Mechanism of spontaneous Heating, factors governing spontaneous heating, methods for study of spontaneous combustion of coal: crossing point method. Stages of spontaneous heating, Detection of spontaneous heating: Physical or sensory indicators, Precautions for prevention of spontaneous heating. Model standing orders in event of fire. Dealing with spontaneous heating & fires: Various methods, Sealing off: selection of site for fire stoppings, construction of fire stopping, fittings to stopping, inspection and maintenance of fire stoppings.

Unit II

Sampling from sealed off area: Sampling line, Methods of sampling from behind the fire stopping, frequency of sampling, Sampling accessible places: water, air displacement method, evacuated, sample, holder, type, Composition and behavior of gases in sealed off area, interpretation regarding condition of heating: Graham's ratio; interpretation regarding explosibility: Cowards Diagram. Elementary knowledge of gas chromatography, Re opening of sealed off area: Factors governing decision of reopening, methods of reopening, selection of methods of reopening, precautions to be taken while reopening. Fire damp explosion: Introduction, Composition of firedamp Modes of emission of Firedamp, Degree of gassiness, methane layering, Mechanism of fire damp explosion, Flammability of firedamp, lower and upper limit of explosibility of firedamp, factors governing limits of flammability, lag on ignition. Explosive limits of other flammable gases, Causes of fire damp explosion and its prevention. Characteristic of firedamp explosion.

Unit III

Coal Dust Explosion: Mechanism of Coal Dust explosion, Flammability limits of coal dust, factors governing explosibility of coal dust, Characteristics of coal dust explosion. Causes of coal dust explosion, Prevention of coal dust explosion. Generalized stone dusting, Quantity of stone dust, Types and properties of stone dust, stone dusting plan. Stone dust barriers, types of stone dust barriers, specifications and construction, location of primary and secondary types of barriers. Situations under which barrier may fail, maintenance and care of stone dust barriers.



Unit IV

Inundation: Surface and underground causes of Inundation and its prevention, water dams, bulkhead doors, Procedure of and precaution while approaching old water logged areas, pattern of bore holes, Dewatering, burn side safety boring apparatus, Standard of lighting in underground & opencast mines, cap lamps, its construction, maintenance and care, cap lamp room. Mine rescue: Introduction, classification of mine rescues apparatus, modern self contained breathing apparatus, its construction, application and scope. Common tests of self contained compressed oxygen breathing apparatus, Chemical oxygen self-rescuers, gas mask, filter self-rescuers: their construction, application and limitations. Fresh air hose type breathing apparatus, Fresh air base: location, personnel & equipments required.

Suggested Reading:

1. *D.J. DESHMUKH VOL I, II*
2. *Mine Environment & Ventilation, G.B. Mishra*
3. *Mine Disaster & Mine Rescue, M.A. Ramlu*

Program: Diploma

Semester: Five

Course: Mine Reclamation and Closure

Course Code: 8D.332

L	T	P	C
3	0	0	3



Course Objective:

- Provide the basis for estimating the financial liability associated with a mining project. The objective of rehabilitating a typical exploration site is to minimize long-term environmental liability by maintaining geotechnical stability, restoring native ecosystems, striving to achieve a more beneficial land use, etc.
- Provide ideas and process about closing a mines, how to do reclamation and necessity of reclamation.

Unit I

Economical and technical aspects of reclamation of mined out land. Reclamation Methods: Back filling, outside dumps and their stability.

Unit II

Top soil handling, assessment of soil productivity potential, re-vegetation, factors for plant Growth, parameters for soil quality and their importance. Reclamation plan and land use plan, general requirements of protection of hydrologic balance.

Unit III

Erosion of soil, types of erosion, estimation of top soil erosion. Landscaping of disturbed land, estimation of reclamation cost and benefits, use of reclaimed land and structures.

Unit IV

Mine Closure Planning: Importance, methodology, statutes concerning mine closure. Land reclamation as post mining operation, Statutes concerning reclamation of mined out area.

Suggested Reading:

1. *Surface Mining Technology*, S.K. Das
2. *Elements of Mining Technology Vol I*, D.J. Deshmukh

Program: Diploma

Semester: Five

Course: Mine Planning & Design

Course Code: 8D.333

L	T	P	C
3	0	0	3

Course Objective:

- Understanding of the role and scope of geomechanics specialists in mining.



- A sound basis for management of the incorporation of geomechanics specialist advice for mine planning and operations is it from company or consultant sources.
- Awareness of the circumstances in which geomechanics specialist advice ought to be sought and applied, and of how that can be engaged and applied.
- The students will have knowledge on planning of opencast mining, underground mining and equipment utilization.

Unit I

Principle of the planning, short range and long range planning, role of planning in mining ventures, ore reserve estimation, economic block model.

Unit II

Mine Planning Input: Geological, mineralogical, structural, economical, environmental and technical inputs. Determination of optimum output, life of a mine and size of mine field based on economic consideration, Taylor's mine life rule, ultimate pit configuration.

Unit III

Optimum location of mine entries, theoretical considerations of opening and development of mine field. Production planning and scheduling, mine equipment planning, estimation of their numbers, infrastructure planning.

Unit IV

Mine Closure-ongoing and final. Feasibility report and project report - contents and preparation.

Suggested Reading:

1. *Coal Mining & Management Vol- I*, S.P. Mathur & N.K. Singh
2. *Modern concept of surface mining*, D. Biswas

Program: Diploma

Semester: Five

Course: Seminar in Executive Communication

Course Code: 40D.401

L	T	P	C
2	0	0	0

Course Objective To impart more advanced basic skills through intensive practice, in this unit again the students get opportunities to apply their general awareness and classroom learning to practical situation to achieve the targeted career goal in this increasingly competitive world Some of the career



oriented units are Discussion Skills, Interview Skills, Job Search Strategies, Job Correspondence etc., they need to undergo,

- An average student acquires basic skills required for a cherished job.
- Their appreciative personality development becomes a value added attribute in their professional sphere.
- The course enhances *communication*, leadership and teamwork *skills*; and personal development *skills* using practical approach and exposure of students to the realities of the world
- To put greater emphasis on development of non-technical skills, such as flexibility, leadership and good *communication*.

Activity Based WORKSHOPS

- Debate
- Extempore
- Group Discussion
- Panel Discussion
- Presentation-Paper & Oral
- Reports: Survey Report, Project Report, Case Study

Suggested Readings:

1. Monippally, Matthukutty. M. 2001. *Business Communication Strategies*. 11th Reprint. Tata McGraw-Hill. New Delhi
2. Swets, Paul. W. 1983. *The Art of Talking So That People Will Listen: Getting Through to Family, Friends and Business Associates*. Prentice Hall Press. New York
3. Lewis, Norman. 1991. *Word Power Made Easy*. Pocket Books
4. Sen, Leena. *Communication Skills*; Eastern Economy Edition
5. Ghanekar, Dr. Anjali. *Essentials of Business Communication Skills*; Everest Publishing House
6. David Green. *Contemporary English Grammar, Structure & Composition*; MacMillan
7. *Dictionary*; Oxford
8. *Dictionary*; Longman

Websites

- www.tatamcgrawhill.com/digital_solutions/monippally
- www.dictionary.cambridge.org
- www.wordsmith.org
- www.edufind.com
- www.english_the_easy_eay.com
- www.englishclub.com
- www.english_grammar_lessons.com
- www.wikipedia.org/wiki/english_grammar



Program: Diploma

Semester: Four

Course: Practical Training Project I

Course Code: 8D.345

L	T	P	C
0	0	2	2

Course Objective:

- Enables the students to experience with the practical applications of the theoretical learning.



- The outcome at the place of work is always much more than what can be learned in the class room.
- To provide the students an opportunity to express their skills, academic knowledge, practical experience and ability to analyze problems.
- The aim of the project is to stimulate creative and innovative aspects of their technological learning.

Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of Semesters.

DIPLOMA IN MINING ENGINEERING												
SEMESTER VI												
S. No	Subject code	Name of Subject	Period			Evaluation Scheme				Subject	Credit	Hours
			L	T	P	Assignment	T A	Total	ESC			
1	8D.351	Mine Legislation & Safety	3	0	0	20	10	30	70	100	3	3
2	8D.352	Fuel Technology & Mineral Processing	3	0	0	20	10	30	70	100	3	3



3	8D.353	Environmental Aspects of Mining	3	0	0	20	10	30	70	100	3	3
4	8D.354	Rock Mechanics	3	0	0	20	10	30	70	100	3	3
5	ELECTIVE II(Any one of the following)											
	8D.381	Mine & mineral Economics	3	0	0	20	10	30	70	100	3	3
	8D.382	Remote Sensing & Geo-informatics										
	8D.383	Engineering Economics										
	8D.384	Engineering Economics										
6	40D.451	** Human Values and Ethics	2	0	0	20	10	30	70	100	0	2
PRACTICAL / SESSIONAL												
1	8DP.352	Fuel Technology & Mineral Processing Lab	0	0	2	-	30	30	50	50	1	2
2	8DP.354	Rock Mechanics Lab	0	0	2	-	30	30	50	50	1	2
3	8D.395	Practical Training Project II	0	0	0	-	-	-		100	3	3
Total										800	20	24

- **** NOTE: Qualifying Non Credit Course & only Viva voce is conducted.**
- **MOOCS introduced through SWAYAM in all semester.**

Program: Diploma

Semester: Six

Course: Mine Legislation and Safety

Course Code: 8D.351

L	T	P	C
3	0	0	3

Course Objective:



- Enables the students to understand the organizational structure, working safety and management concepts for mining enterprises to effectively manage them within the frame work of rules and regulations.
- Students will be acquainted with rules, laws and order for running a mine.
- The students will have knowledge on various acts, rules and regulations relating to the mineral industry. They will also know about accidents, diseases and mine safety.
- Student will know about subsidence control, governing norms and regulations.

Unit I

Mines Act: Important definition: Adolescent, adult, child, Employed, Mine, Open cast working, Relay, Shift, Serious bodily injury. Provisions under chapter V, Provision for health and safety. Provisions regarding leave with wages, Act 49 to 56 .Hours & Limitations of Employment, act 28 to 48.Mines rules: Provisions regarding health & sanitation, first aid and medical appliances. Mines Rules- Provisions connected with leave with wages and over time and welfare amenities. Employment of persons, Rule 46 to 52

Unit II

Coal Mines Regulations: Important definitions: Duties and responsibilities of workman, competent person & officials. Provisions of Reg. 38, 39, 43, 44, 45, 46, 48, 56 Planes and sections Reg. 58, 59, 61, 63.Means of access & egress. Reg. 66 to 70 Provisions regarding winding in shaft Reg. 71 to 86. Transport of men & material Reg. 88, 89,90,91,92,93,94,95 mine working Reg. to 115 Precautions against dangers from the dust, gas & water .Reg. 116 to 128. Ventilation Reg. 130 to 149 Provisions regarding lighting and safety lamp. Reg. 150 to 158. Explosives & Blasting Reg. 158 to 180 Provisions regarding machinery, plant & equipments and important provisions under chapter on miscellaneous.

Unit III

Mine accidents: Types of mine accidents, their classifications, Causes of accidents due to fall of roof, explosives and blasting, haulage and winding and their preventions, Cause and prevention of accidents due to, fires, explosions and inundations. Safety statistics, safety drive and organization of safety in the mines/area etc. Management: Types of business organizations, organization of Coal India Ltd. Supervision qualities of good supervisors, Leadership, functions of industrial leadership, delegation of responsibility , Principles of time study, Wage and payment , Trade unions, their functions, Strikes and lockouts.

Unit IV

Circulars, Byelaws & standing orders: Model standing order in the event of stoppage of main mechanical ventilator, Maximum air velocity, Systematic support rules for coal mine with Board and pillar method of working, Conditions for solid blasting with P5 explosives, Precautions for use of Auxiliary fan underground. Procedure for dealing with misfire and Precautions regarding Blown through shots. Inspection Procedure: Procedure of inspection of old working, Haulage roadways, sinking shaft, working shaft, Winding rope, Sealed off area, subsidence and goaf area Mines Rules: Important provisions of coalmines rescue rules: Organization & equipment in mines, Rescue station, Conduct of rescue work.

Suggested Reading:



1. *Mines Act, 1952, Lovely Prakashan*
2. *CMR, 1957, L C Kaku*

Program: Diploma

Semester: Six

Course: Fuel Technology and Mineral Processing

Course Code: 8D.352

L	T	P	C
3	0	0	3

Course Objective:



- The course enables the students to select the suitable parameters and appropriate machinery for processing various types of minerals.
- Give the students an understanding of how the basic mineral processing unit operation can be combined into specific processes.
- Students will be acquainted with mineral separation methods – an overview, including: physical separation (gravity, magnetic and electrical); solid/liquid separation; flotation.

Unit I

Mine sampling: Definition, terms, purpose and various uses. Different Sampling methods. Salting- purpose, safety against salting. Reduction of sampling-Methods used.

Unit II

Assaying: Introduction- assay map, assay plan factor, assay values, grade value, tenor, type of grade value. Calculation based on average assay value. Estimation of ore reserves.

Unit III

Mineral dressing: Scope, objectives & limitations of mineral dressing. Comminution. Size separation. Gravity concentration methods. Introductory froth flotation. Simplified flow sheets of coal, copper, Lead, & Zinc, Iron, Limestone, Coal Beneficiation methods

Unit IV

Fuel technology: Proximity & ultimate analysis of coal, caking & coking properties of coal, low & high temperature carbonization.

Suggested Reading:

1. *Mineral Processing Technology –Barry A. Wills*
2. *Basics in Mineral Processing- Mesto*
3. *Mineral Processing Technology- Gaudin*

Program: Diploma

Semester: Six

Course: Fuel Technology and Mineral Processing Lab

L	T	P	C
0	0	2	1



Course Code: 8DP.352

List of Experiments:

1. Proximity analysis of coal.
2. Caking index of coal.
3. Crushing by Jaw roll crusher.
4. Grading of iron ore fines.
5. Floating of coal fines.

Program: Diploma

Semester: Six

Course: Environmental Aspects of Mining

Course Code: 8D.353

L	T	P	C
3	0	0	3



Course Objective:

- Enable the students to understand the ill effect of mining activities on environment and how to reduce it.
- Upon completion of this course, the students can able to identify the new methodologies / technologies for effective utilization of renewable energy sources.

Unit I

Environmental aspects of Mining & associated Activities:

Ecosystem structure and function, Effects on Biodiversity due to mining, Effects on water bodies, subsidence and its effect.

Unit II

Pollution and its effects: Air pollution: sources, and

control measures, Noise and vibration: Sources and control measures. Water pollution: Preventive and control measures, acid rain. Soil pollution: its effect, prevention and control measures.

Unit III

Reclamation: Land Degradation due to mining & its reclamation.

Various methods of reclamation and process.

Unit IV

Resource depletion: Exploitation of natural resources, over

drafting, overexploitation, Environmental impact of the coal industry, exporting of hazardous waste

Suggested Reading:

1. *Mining and its Environmental Impact- R E Hester, R M Harison*
2. *Mining Environmental handbook- Jerrold J Marcus*

Program: Diploma

Semester: Six

Course: Rock Mechanics

Course Code: 8D.354

L	T	P	C
3	0	0	3



Course Objective:

- The course provides detailed knowledge on rock properties
- This will equip the students with the ability to carry out various tests and monitoring the rock behavior.
- Students will be able in analysis of analysis of data and solving rock mechanics problem in mining and excavation projects.
- Data and solving rock mechanics problem in mining and excavation projects.
- Provides detailed knowledge on rock properties and equips the students with the ability to carry out various tests.
- Students will be able in analyzing the data and solving rock mechanics problem in mining and excavation projects.
- The students will have knowledge about the subsidence mechanism, prediction and influencing parameters.
- The students will have the concept about the rock mass classification, mechanism of rock reinforcement, existing and special methods of rock reinforcement.

Unit I

Introduction to Rock Mechanics: Concept of stress and strain in rock, stress due to weight of strata, vertical lateral stresses. Stress due to tectonic and orogenic force, Residual stresses, induced stresses. Field stresses, modulus of elasticity poisson's number, Poisson's ratio stress fields. Introduction to elementary rock mass classification based on strength, hardness, RQD, Bieniawski RMR classification.

Unit II

Rock Mass Properties: Strength Properties: Compressive strength, Tensile Strength, Shear Strength, Flexural Strength. Strength Indices- Point Load Strength index, Impact Strength index, Protodyakonov strength index. Rebound hardness, insitu stress by flat jack, Cohesion, Young's modulus, poisson's ratio, angle of internal friction. Porosity, Density, Moisture content permeability. Material Characteristics: Brittle material, ductile material, Elastic material, Plastic material. Time dependent properties: creep. Creep curve, factors contributing Creep, deformation.

Unit III

Rock Testing: Uniaxial compressive strength, Tensile strength – Brazilian test, bending test. Shear strength test- punch shear test, direct shear test on Rock cube, Triaxial method. Determination of strength indices- point load strength index, Protodyakonov strength index, impact strength index. Rock burst, Bumps, causes controlling measures, factors affecting proneness to rock burst/Bumps. Pillar Design- factors considered. Pillar design by tributary area approach, determination of factor of safety.

Unit IV

Ground control: Theories of mechanics of strata behavior: Dome or arch theory, Beam theory, Function of roof bolts, Principle of Action Roof Bolts, Varieties of Roof Bolts: Slot and Wedge, Expansion shell, Grouted Roof Bolts and Resin Roof Bolts, Anchorage Testing of Roof Bolts, Bolt density, Code of practice for roof bolting in underground mines, Roof stitching, Principle of Roof stitching, Cable Bolting.

Suggested Reading:

1. *Elements of Mining Technology Vol I*, D.J. Deshmukh
2. *The elements of mechanics of mining ground*, B.S. Verma
3. *Rock Mechanics for Engineers*, Dr. B.P. Verma





Program: Diploma
Semester: Six
Course: Rock Mechanics Lab
Course Code: 8DP.354

L	T	P	C
0	0	2	1

List of Experiments:

1. Preparation of rock sample for laboratory testing.
2. Determination of Uniaxial compressive strength of a rock sample.
3. Determination of tensile strength (Brazilian test) of a rock sample.
4. Determination of shear strength of a rock sample.
5. Determination of point load strength index.
6. Determination of Protodyakonov strength index.
7. Determination of impact strength index.
8. Demonstration of various Rock bolts.



ELECTIVE -II

Program: Diploma

Semester: Six

Course: Mine and Minerals Economics

Course Code: 8D.381

L	T	P	C
3	0	0	3

Course Objective:

- The course enables the students to understand the economics of business enterprise to become a successful manager.
- Students gain knowledge on the basic management principles to become management(s) professional.
- Upon completion of the course, students will be able to gain knowledge and skills Needed to run a business successfully.
- Expertise the students to brings employment, government revenues, and opportunities for economic growth and diversification through mining.
- Study of estimation and valuation of mineral deposit and study of project appraisal.

Unit I

Mineral Industry: Mineral Industries in India, Role of Mineral Industries in National Economy. Major Economical minerals coal, Iron, Copper, Manganese, Limestone, Lead and Zinc, Gold, Radioactive minerals. Geological formation modes, Locations, Reserve, Uses, Production, Imports, Exports. Conservation of Minerals and their substitution including coal. National Mineral Policy. Incentives provided by government to Mining sector, Computation & classification of Reserve and Grades.

Unit II

Valuation & depreciation: Valuation methods of valuation by different methods of annuity. Calculation of different annuities, Methods of depreciation and calculation of Depreciation methods of calculations of Redemption values. Main valuation methods of mining Property which under production. Valuation under different methods. Report of valuation of small mining property.

Unit III

Mine Leasing Procedure: Different Acts, Rules related to Scientific Development, Regulation and Conservation of Minerals, major and minor mineral concessions. General Restriction on undertaking mining operation, Maximum area for which Mining lease may be granted, Period for which mining lease may be granted, procedure for obtaining mining lease.

Unit IV

Mining Plan: As per the MCDR, 1988: MCR, 1960. Procedure of arranging finance for small-scale mining through financial institutes, Mine Closure plan.



Suggested Reading:

1. *Mine Valuation, sparks*
2. *Mine Economics, R.T. Deshmukh*
3. *Elements of Mineral exploration, IBM*

Program: Diploma

Semester: Six

Course: Remote Sensing & Geo-Informatics

Course Code: 8D.382

L	T	P	C
3	0	0	3

Course Objective:

- Student will understand potential application of remote sensing/GIS applications
- Enable students to learn the background to general image processing and GIS operation in order to extract and manage spatial information
- Students will Understand advanced surveying techniques such as remote sensing and geodetic surveying

Unit I

Basic concepts of remote sensing, airborne and space borne sensors, present status of remote Sensing satellites, data acquisition techniques from different sources. Digital image processing, restoration, image enhancement,

Unit II

Segmentation feature extraction, Clustering edge detection, introduction to digital terrain modeling. Geographic Information System, introduction to microwave remote sensing and Global Positioning System.

Unit III

Application of GPS in remote sensing. Application of remote sensing. Applications of remote sensing in mineral resource identification and estimation, mine environment and ecology, post mining land reclamation, mine disaster management and reuse of mined out area

Unit IV

Use of relevant software's for remote sensing and GIS with particular reference to mining. Hands on exercises on image processing and GIS packages

Suggested Reading:

1. *Surveying and leveling Vol. I & II , T. P. Kanetkar*
2. *Surveying & Levelling , B.C. Punmia*
3. *Surveying & Levelling , Amarjit Aggarwal*
4. *Remote Sensing & GIS , Basudeb*



Program: Diploma

Semester: Six

Course: Engineering Economics

Course Code: 8D.383

L	T	P	C
3	0	0	3

Course Objective:

- Enable students to perform and evaluate present worth, future worth and annual worth analyses on one of more economic alternatives.
- Students will have knowledge of liability, insurance and safety/environmental requirements on projects.
- Prepare engineering students to analyze cost/revenue data and carry out make economic analyses in the decision making process to justify or reject.
- The objectives of this course are to review with students basic economic principles with students.
- Educate the students on how to systematically evaluate the various cost elements of a typical manufactured product, an engineering project or service, with a view to determining the price offer.

Unit I



Introduction, demand and supply, statistical demand and supply analysis, managerial application of the elasticity of demand and supply.

Unit II

Estimation of production and cost function; forecasting techniques and their use for decision making at different levels; linear programming and production analysis. Price output determination: a review with special reference to condition of uncertainty, kinds of pricing problems.

Unit III

Capital market and investment decisions: pay back method, average return on investment or accounting rate of return, net present value, and internal rate of return.

Unit IV

Outline of welfare economics, resource accounting and sustainability, income determination and fluctuations.

Suggested Reading:

1. *Engineering Economics*, J A L Waddell
2. *Engineering Economics*, R pannerselvam

Program: Diploma

Semester: Six

Course: Principle of Management

Course Code: 8D.384

L	T	P	C
3	0	0	3

Course Objective:

- To study the rudiments of business and various types of business.
- To understand the intellectual property rights.
- To comprehend the management process, organization and different functions of management.
- To perceive the principles of human resource management and financial management.

Unit I



Overview of Business: Types of Business, Service, Manufacturing, Trade, Industrial sectors, Introduction to Engineering Industry, Process Industry, Textile Industry, Chemical Industry. Agro Industry, Globalization, Introduction, Advantages & disadvantages w.r.t India, Intellectual Property Rights I (I P R), Concept, Types of IPR.

Unit II

Management Process: What is Management? Evolution, Various Definitions, concept of management, Levels of Management, Administration and Management, Functions of management: Planning, Organizing, coordinating, Directing, controlling, decision making.

Organizational Management: Organization, Definition, Steps in forming organization, Types of Organization, Line, Line & Staff, Authority & Responsibility, Span of Control (Management).

Unit III

Human Resource Management: Personnel Management: Introduction, Definition & Function Staffing: Introduction to HR, Introduction to HR Planning, Recruitment procedure. Personnel – Training & Development: Types of training- Induction- Skill enhancement. Safety Management: Causes of Accidents, Safety Procedures.

Unit IV

Financial Management: Financial Management- Objectives & Functions, Capital Generation & Management, Types of capitals, Sources of finance.

Suggested Reading:

1. *Essentials of Management*, Harold Koontz & Weirich: (Tata McGraw Hill)
2. *Principles & Practices of Management*, L.M. Prasad, S.Chand
3. *Management*, Stephen Robbins, INS P

Program: Diploma

Semester: Six

Course: Human Values and Ethics

Course Code: 40D.451

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Course Objective:

- To develop students' sensibility with regard to issues of gender in contemporary India.
- To provide a critical perspective on the socialization of human beings.
- To introduce students to information about some key aspects of Indian culture and ethics.
- To expose the students to debates on the politics and economics of work.
- To help students reflect critically on gender violence.
- To expose students to more egalitarian interactions between men and women.



- Students will attain a finer grasp of how gender discrimination works in our society and how to counter it.

Unit I

1. VALUE CRISIS IN CONTEMPORARY INDIAN SOCIETY

- 1.1 Value Crisis at the Individual Level
- 1.2 Societal Level
- 1.3 Intellectual Level
- 1.4 Cultural Level
- 1.5 Value – What are they?
- 1.6 The Indian Concept of Values.
- 1.7 Modern Approach to the Study of Values.
- 1.8 Aesthetic Sensibilities

Unit II

2. MORAL AND ETHICAL HUMAN VALUES

- 2.1 Bases for Moral Judgment
- 2.2 Some Canons of Ethics.
- 2.3 Virtue Ethics.
- 2.4 Ethics of Duty.
- 2.5 Ethics of Responsibility
- 2.6 Factors to be considered in Making Ethical Judgments.
- 2.7 Different Meanings of Human Values
- 2.8 A New Approach to Human Value ,Freedom, Creativity Love &Wisdom

Unit III

3. MORAL VALUES IN PROFESSION

- 3.1 What is a Profession?
- 3.2 Professional Ethos
- 3.3 Code of Professional Ethics
- 3.4 Practicing the Code
- 3.5 Corporate Social Responsibility
- 3.6 The Larger Domain of Human Values
- 3.7 Institutionalizing Ethics and Human Values

Unit IV

4. GENDER SENSITIZATION

- 4.1 Socialisation of women
- 4.2 Just Relationships, being together as equals
- 4.3 Declining sex ratio, demographic consequences
- 4.4 Women's work, its politics and economics, fact and fiction, unrecognized and unaccounted work
- 4.5 Domestic violence, eve teasing and harassment. Is home a safe place?

Recommended Texts:



1. Dr. Rajan Mishra, *Human Values*: Laxmi Publications Pvt. Ltd.
2. S. Dinesh Babu, *Professional Ethics and Human Values*; Laxmi Publications Pvt. Ltd.
3. P.S. Rathore. *Business Ethics And Communication*; S.Chand Publishing
4. Dr. K.Alex. *Managerial Skills*; S. Chand Publishing.
5. Dr. M. Adithan, *Study Skills For Professional Students For Higher Education* , S.Chand Publishing
6. Govindarajan M “*Professional Ethics and Human Values.*”
7. R.R. Gaur and R. Sangal “*A Foundation Course in Human Values and Professional Ethics*”

Websites:

- [www.tatamcgrawhill.com/digital Solutions/](http://www.tatamcgrawhill.com/digital%20Solutions/) monopoly
- www.schandedutech.com
- www.laxmipublications.com

Program: Diploma

Semester: Six

Course: Practical Training Project II

Course Code: 8D.395

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Course Objective:



- Enables the students to experience with the practical applications of the theoretical learning.
- The outcome at the place of work is always much more than what can be learned in the class room.
- To provide the students an opportunity to express their skills, academic knowledge, practical experience and ability to analyze problems.
- The aim of the project is to stimulate creative and innovative aspects of their technological learning.

Vocational Training in a Surface / Underground Mine of minimum 30 (Thirty) days to be taken at the end of Semesters.